# 1 Acute and Chronic Inflammation

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## INTRODUCTION

The inflammatory response consists of an innate system of cellular and humoral responses following injury (such as after heat or cold exposure, ischemia/reperfusion, blunt trauma, etc.), in which the body attempts to restore the tissue to its preinjury state. In the *acute inflammatory response*, there is a complex orchestration of events involving leakage of water, salt, and proteins from the vascular compartment; activation of endothelial cells; adhesive interactions between leukocytes and the vascular endothelium; recruitment of leukocytes; activation of tissue macrophages; activation of platelets and their aggregation; activation of the complement; clotting and fibrinolytic systems; and release of proteases and oxidants from phagocytic cells, all of which may assist in coping with the state of injury. Whether due to physical or chemical causes, infectious organisms, or any number of other reasons that damage tissues, the earliest in vivo hallmark of the acute inflammatory response is the adhesion of neutrophils (polymorphonuclear leukocytes, PMNs) to the vascular endothelium ("margination") (Figure 1.1). The *chronic inflammatory response* is defined according to the nature of the inflammatory cells appearing in tissues. The definition of chronic inflammation is *not* related to the duration of the inflammatory response. Reversal or *resolution* of the inflammatory response implies that leukocytes will be removed either via lymphatics or by apoptosis (programmed cell suicide) and that the ongoing acute inflammatory response is terminated. As a consequence, during resolution increased vascular permeability is reversed due to closure of the open tight junctions and PMN emigration from the blood compartment ceases. In both the vascular and extravascular compartments, fibrin deposits are removed by pathways that lead to activation of plasminogen (to plasmin), which degrades fibrin (see section on "Intercommunications between Inflammatory Cascades and the Coagulation Cascade"). Cell debris and red blood cells (RBCs) in the extravascular compartment are removed by phagocytosis involving tissue macrophages. There are many situations in which

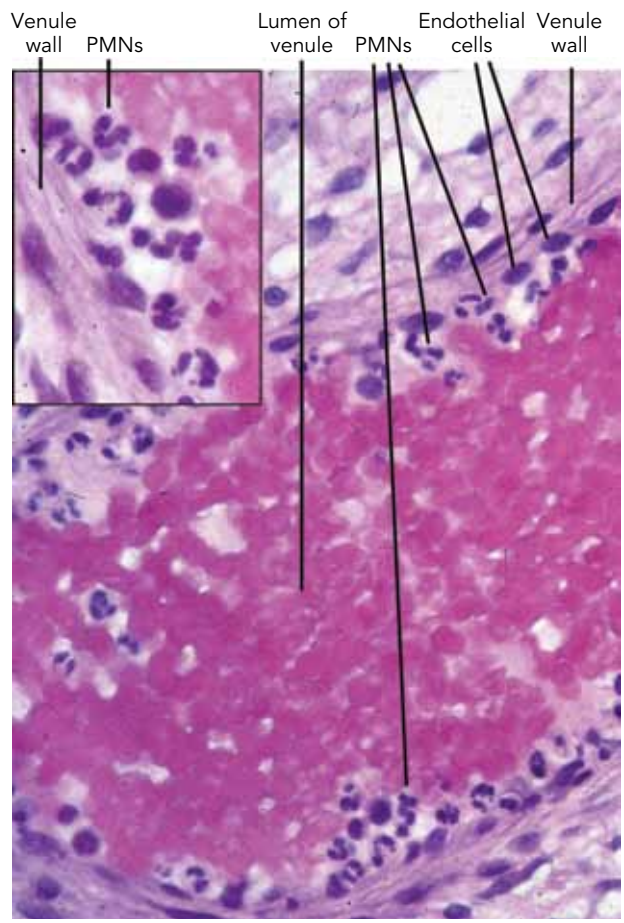



**Figure 1.1.** Early changes in a human venule involved in an acute inflammatory response. PMNs are "marginating" along the endothelial surface of a venule preparatory to their migrating into the extravascular space due to their adhesion to endothelial surfaces.



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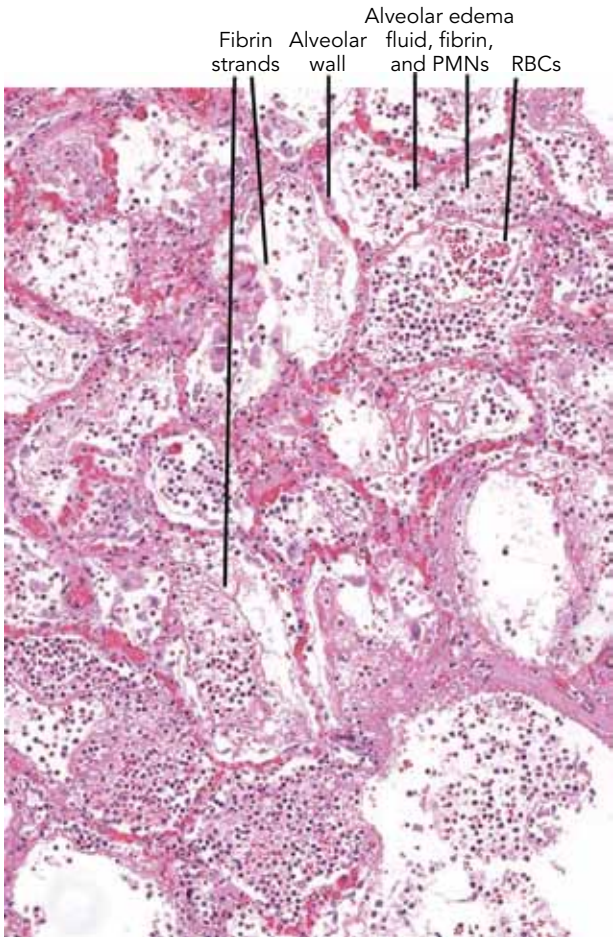



**Figure 1.2.** Acute bacterial pneumonia (*Streptococcus pneumoniae*) in human lung. Most alveolar spaces are filled with PMNs, fibrin strands, edema fluid, and RBCs, which clinically results in serious interference with gas exchange between the air spaces and vascular compartment.

the acute inflammatory response becomes excessive or prolonged, leading to serious damage of tissues and organs. Examples of unremitting acute inflammatory responses resulting in injury are discussed in the following section. Presented subsequently are concepts regarding acute and chronic inflammation. These are designed to provide the reader with a conceptual framework for an understanding of the inflammatory responses, their causation, and the outcomes.

## THE ACUTE INFLAMMATORY RESPONSE

The acute inflammatory response is defined as a series of tissue responses that can occur within the first few hours following injury. In cases of bacterial pneumonia (Figure 1.2), bacterial meningitis (Figure 1.3), or ischemic myocardial injury (Figure 1.4), the inflammatory response is exuberant within the first few hours or days and then gradually declines unless the offending agent (such as bacteria-inducing pneumonia) cannot be cleared by phagocytosis. Resolution of the inflammatory response requires killing of bacteria and removal of their debris. When tissue injury occurs in various organs, the resolution of the inflammatory response may, to an extent, involve regeneration in which an organ can rapidly replace damaged or destroyed cells with an architectural outcome that resembles the original uninjured tissue. A good example of this is acute injury (chemical, viral, etc.) in the liver in which the inflammatory response resolves via regeneration of liver cells (hepatocytes, Kupffer cells, endothelial cells, etc.), reconstituting the damaged or destroyed liver, with the end result being tissue that is virtually identical to that before injury. In other situations, such as in the myocardium, ischemic destruction of cardiomyocytes results in an intense acute inflammatory response with a heavy build-up of PMNs (Figure 1.4). The regenerative abilities of the myocardium are extremely limited, if at all, and the destroyed myocardium must be quickly replaced with a fibrous (collagenous) scar. If scar formation is insufficient, the clinical outcome may be fatal due to cardiac rupture and a filling of the pericardial cavity with blood (hemopericardium). In the central nervous system, the commonly held belief is that vital cells such as neurons are largely unable

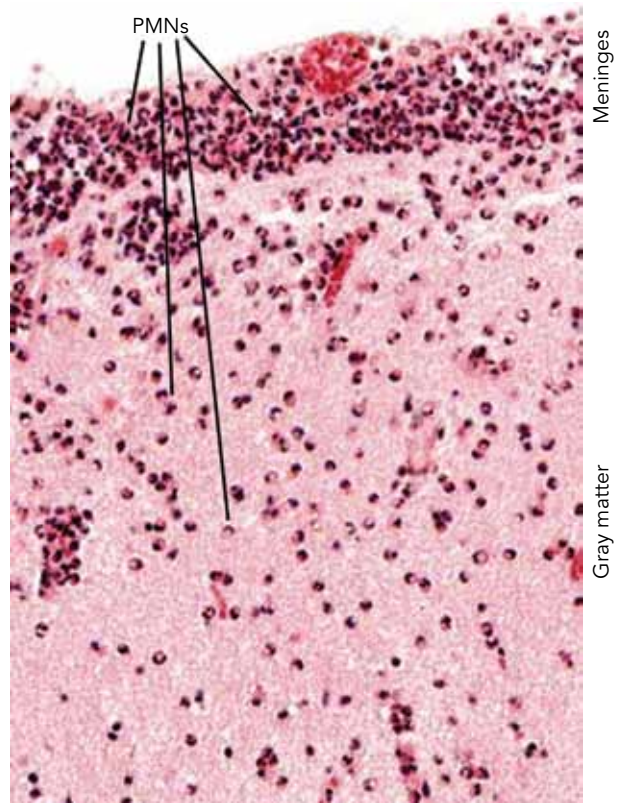



**Figure 1.3.** Acute bacterial meningitis (*Neisseria meningitidis*) with an intense accumulation of PMNs in the brain covering (meninges) and scattered accumulations of PMNs in the gray matter of the brain.

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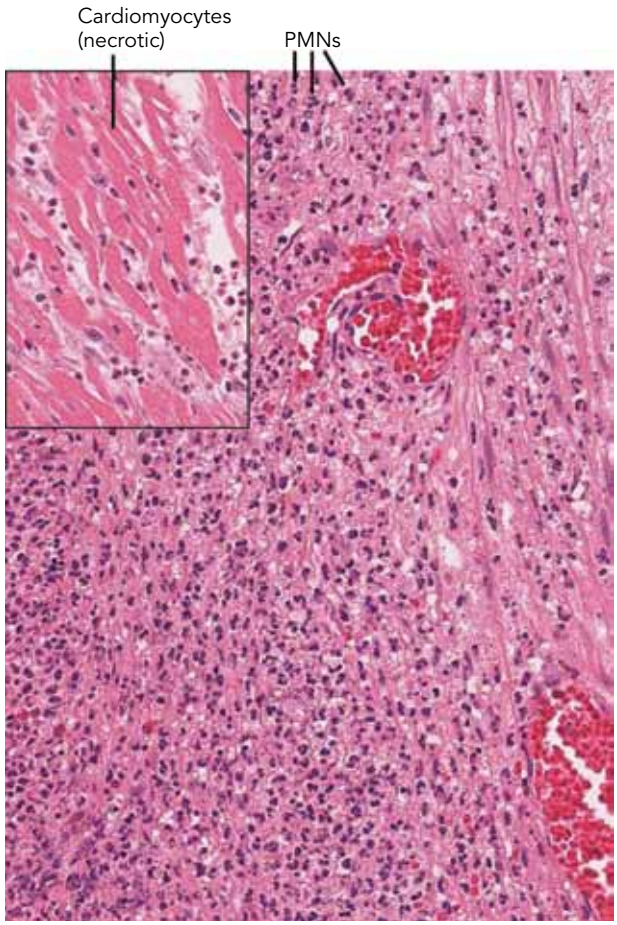



**Figure 1.4.** An acute inflammatory response in the myocardium 48 h after clinical onset of acute ischemic injury. The main frame shows an intense PMN infiltrate. The inset shows the edge of the infarct area with a necrotic area of myocardium and interstitial accumulation of PMNs.

to regenerate so that, once lost, replacement cannot occur. Most of the theories about cellular regeneration are undergoing reconsideration in the face of extensive stem cell work, which raises the question as to whether locally present or circulating stem cells may become actively involved in replacing cells similar to those that have been destroyed. This is an intensely debated topic that requires much additional investigation.

Figure 1.5 describes changes in tissues related to induction of the acute inflammatory response regardless of what has incited this rejection. There are numerous changes within the vascular compartment that trigger the acute inflammatory response, involving at least six intravascular events. The first change is *activation of endothelial cells*, during which these cells begin to express on their surfaces adhesion molecules for leukocytes (see Chapter 18 for detailed information about these molecules and their "counter-receptors" on leukocytes). Activated endothelial cells also engage in the generation and release of proinflammatory cytokines and chemokines (Chapter 13), which will chemotactically attract and activate PMNs as these cells adhere to the endothelium before their transmigration into the extravascular compartment. Activated endothelial cells often also express tissue factor (TF, aka Factor III) on their luminal surfaces. The complement activation product, C5a, can cause upregulation of TF on endothelial cells. TF is a potent procoagulant that can lead to thrombus formation on the vascular surface. A second event is *reversible opening of endothelial cells tight junctions*, which allows for the leak of protein and fluids from the vascular compartment into the extravascular compartment. When extensive edema develops in closed compartments, such as in the central nervous system (during bacterial meningitis) or in articular joints (after trauma), this can result in increased hydrostatic pressure that can seriously impair organ function. In the lung, extensive edema formation in the alveolar compartment (also known as "alveolar flooding") can seriously compromise air exchange between the alveolar and vascular compartments. Examples of this problem occur in acute high-altitude sickness, during bacterial pneumonia, and during Hanta virus infection of the lung. An inability for adequate gas exchange between

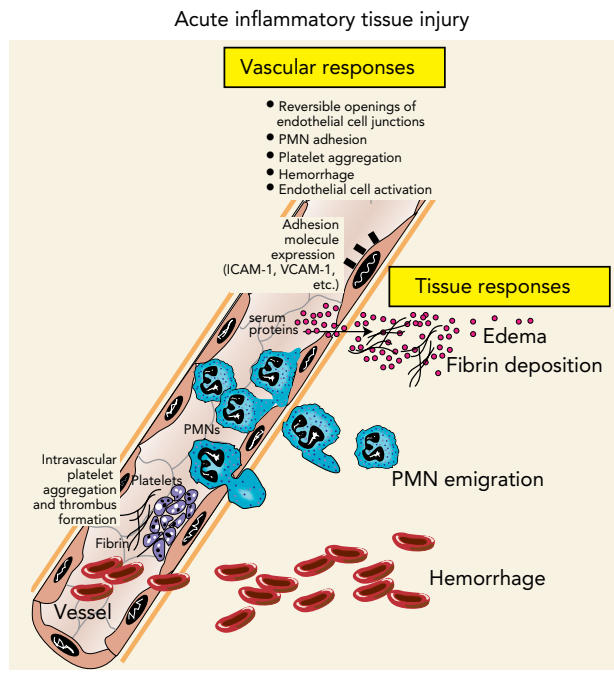



**Figure 1.5.** Mechanisms of acute inflammatory responses, featuring the initial intravascular events that lead to increased permeability changes, activation of endothelial cells (increased expression of adhesion molecules for neutrophils [PMNs]), adhesion of PMNs to the endothelial surfaces, and platelet activation (resulting in their aggregation and adhesion to one another as well as to endothelial surfaces). Platelets are often present in areas of fibrin deposition. Tissue responses (in extravascular compartment) feature edema, transmigration of PMNs, fibrin deposition, and hemorrhage if the structural integrity of the vascular barrier has been compromised.

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**TABLE 1.1. Factors affecting vascular integrity**

| | Events | Factors responsible |
|---|---|---|
| Edema | Reversible opening of endothelial tight junctions | Histamine, serotonin, kinins (bradykinin), C3a, C5a, •NO, PAF, prostaglandins |
| PMN emigration | Movement beyond vascular barrier | Chemoattractants: C5a, cytokines (IL-1β, TNFα), CXC, chemokines: collagen and bacterial peptides, metabolites of arachidonic acid |
| Hemorrhage | RBCs in extravascular compartment | Physical forces (heat, cold), bacterial products, proteins and oxidants from phagocytes |
| Platelets | Intravascular aggregation and fibrin formation | PAF, ADP, thrombin activation, etc. |

the alveolar and vascular compartments can lead to a life-threatening state of hypoxia requiring intensive resuscitative support. Reversibility of the junctional changes in the vascular endothelium implies that open endothelial junctions can close to contain the amount of edema fluid accumulating in the extravascular compartment. Agents such as histamine are well known to interact with the vascular wall to bring about reversible opening of the tight endothelial junctions (Table 11.1 of Chapter 11). Whether vasopermeability mediators directly affect vascular endothelial cells (resulting in their contractility that opens tight junctions) or whether there are periendothelial cells which are tethered to endothelial cells (that respond by contraction to factors such as histamine, pulling the tight junctions open) is a matter of considerable debate. Another key factor in the acute inflammatory response is adhesive interactions between PMNs and endothelial cells (Chapters 4 and 11). Ordinarily, PMNs and other leukocytes are carried in the center in the blood stream without making contact with endothelial surfaces. PMNs undergo activation responses such as upregulation of CD11b/CD18 on their cell membranes (Chapter 4), while endothelial cells undergo activation most commonly with gene expression, leading to appearance of adhesion molecules on the laminar faces of endothelial cells. Examples of these molecules are P-selectin, E-selectin, and ICAM-1 (Chapter 18). The sequence of intermittent PMN adhesion to the endothelium (described as PMN rolling) followed by tight adhesion and eventual transmigration of PMNs through endothelial cell junctions is described in detail (Chapter 18). It should be pointed out that studies featuring intermittent ("rolling") followed by firm adhesion have been focused on changes in postcapillary venules. In the case of the lung, PMN transmigration occurs in capillaries that would not permit the rolling phenomenon described above because of physical constraints (inadequate space, since PMNs have a diameter equivalent to the diameter of a capillary). Furthermore, as the inflammatory response commences, PMNs become activated and stiff and cannot undergo deformity to adjust to the tight confines of capillaries. Nevertheless, vascular adhesion molecules play an important role in the transmigration of PMNs in the lung, since the absence or blockade of adhesion molecules clearly diminishes the build-up of PMNs in the extravascular space. Another feature of the acute inflammatory response is platelet activation, which is usually associated with the conversion of prothrombin to thrombin. Platelets can also be directly activated by various other agents (Table 1.1). The end result is platelet adhesion to one another (resulting in platelet aggregates) as well as to endothelial cells. This is the forerunner of intravascular thrombosis in which fibrin deposition develops in and around aggregated platelets. Finally, the acute inflammatory response may be associated with hemorrhage because of direct structural damage (reversible or irreversible) to the endothelial barrier. The development of hemorrhage implies that the vasculature has been severely damaged, since RBCs have no intrinsic mobility and are passively carried out of the vasculature if there has been sufficient loss of vascular integrity. Hemorrhage occurs after thermal or cold trauma, in situations of severe platelet dysfunction or platelet deficiency, after infections due to the release of toxins (as from Streptococcal A bacteria), and in patients undergoing excessive anticoagulant therapy, to cite a few examples. As indicated earlier, all of these changes of the acute inflammatory response are reversible. Edema fluid is cleared from the distal airway (alveolar compartment) by uptake of these fluids together with inflammatory cells into the draining lymphatics, with return to the blood compartment. Thus, what comes from the blood compartment often returns to the blood compartment. PMN clearance in tissues may also occur by apoptosis of these cells and their phagocytosis by tissue macrophages. Thrombosis within vessels can be cleared by activation of the fibrinolytic system, involving tissue plasminogen activator (TPA) and other factors that will activate the fibrinolytic enzyme, plasmin (Figure 1.6).

Table 1.1 lists factors that affect the vascular integrity and lead to changes in the endothelium, resulting

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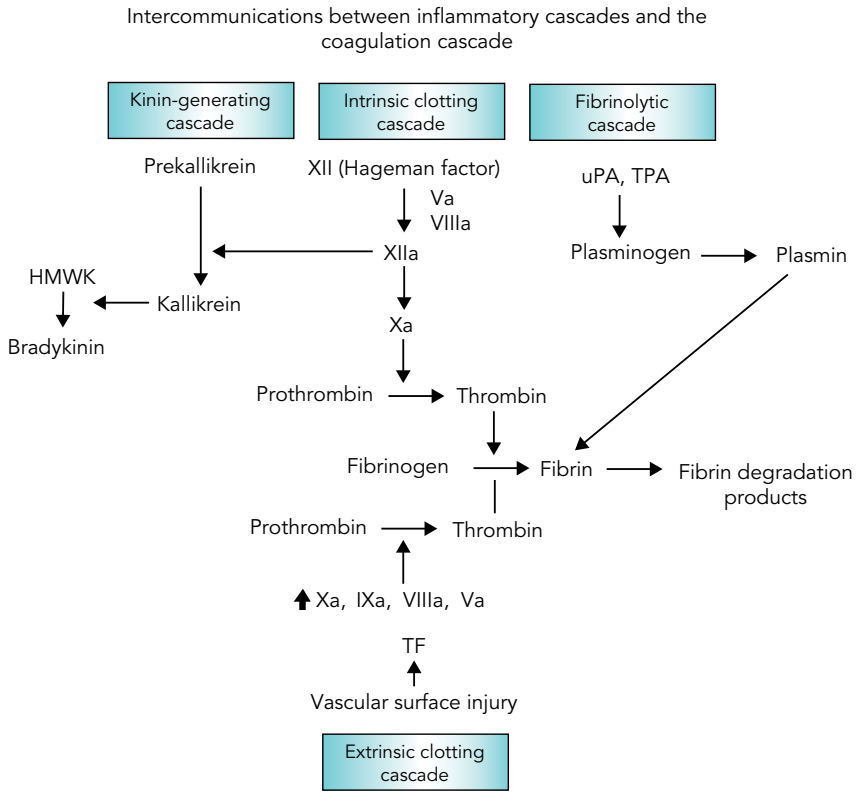



**Figure 1.6.** The coagulation cascades (intrinsic and extrinsic) and intercommunications with kinin generating and fibrinolytic cascades.

in edema formation, PMN accumulation, platelet activation, and development of hemorrhage. Edema due to reversible openings of endothelial cell tight junctions can be induced by histamine; serotonin; kinins (such as bradykinin); the complement anaphylatoxins (C3a and C5a), which act on mast cells to release histamine; nitric oxide; platelet activating factor (PAF); and certain prostaglandins (Chapter 12). As indicated earlier, these responses resulting in edema fluid accumulation outside the vascular compartment are usually reversible and transient. PMN emigration is preceded by adhesive interactions between these cells and endothelial cells via engagement of adhesion molecules on both cell types (e.g., E- and P-selectins on endothelial cells; CD11b/CD18 on PMNs, etc.) (as described earlier and in Chapter 18). Ordinarily, adhesion molecules are present in low quantities on endothelial and PMN surfaces but activation of either cell type can dramatically and rapidly or slowly increase the levels of adhesion molecules that appear either following fusion of cytosolic granules to the cell membrane in the case of PMNs (the rapid response occurring within minutes) or following transcriptional upregulation (a slow response requiring hours). Adhesion molecules on endothelial cells include ICAM-1 and E-selectin, although in the case of P-selectin, this adhesion molecule can be upregulated either via transcriptionally dependent responses or via transcriptionally independent responses. Rapid expression is due to P-selectin addition to the cell membrane via fusion of cytosolic granules (Weibel–Palade granules). Rapid upregulation of adhesion molecule is found in PMNs, platelets, and endothelial cells. In the case of PMNs, increased expression of cell membrane adhesion molecules (CD11b/CD18) is usually rapid due to fusion of secondary granules in the cytosol (which contain adhesion molecules on their inner surfaces) to the cell membranes of PMNs. Chemoattractants for PMNs responsible for their extravascular migration (emigration) include C5a, cytokines (such as IL-1β and TNFα), CXC chemokines, collagen and bacterial peptides, as well as metabolites of arachidonic acid, all of which are described in Chapters 4, 12, and 13. Proteases and oxidants from activated phagocytic cells cause damage to the endothelial barrier, as well as to cells and connective tissue matrix, resulting in widespread damage of both the vascular and extravascular compartments (Chapter 17). Platelets can be activated by a variety of factors (Chapters 5, 8, 12, and 15), such as PAF, ADP, and thrombin, which is activated when the clotting cascade has been triggered (see section on "Intercommunications between Inflammatory Cascades and the Coagulation Cascade").

As suggested in the earlier comments, the acute inflammatory response is a protective shield against

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**TABLE 1.2. Regulation of the acute inflammatory response: natural anti-inflammatory factors**

| | Factors | Targets |
|---|---|---|
| Cytokines | IL-4, IL-10, IL-12 | • Stabilization of IκB and reduced NF-κB activation |
| Protease inhibitors | SLPI, TIMP-1, $\alpha_1$PI, etc. | • Inhibition of serine proteases and nonserine proteases |
| Antioxidant enzymes | Superoxide dismutase<br>Catalase,<br>Glutathione peroxidase | • Converts $\bullet O_2$ to $H_2O_2$<br>• Destroys $H_2O_2$<br>• Catalyzes the breakdown of $H_2O_2$ to $H_2O$ |
| Lipoxins | | see Chapter 12 |
| Glucocorticoids | | • Diverse |
| Kinases | Hydrolysis of kinins | • Bradykinin, etc. |
| Phosphatases | Removal of phosphates from proteins | • Transcriptional factors |
| Transcriptional factors | STAT3, SOCS3 | • Blockade of gene activation for proinflammatory mediators |

tissue that has been damaged. The purpose of the response is to return the tissue to its predamaged state. In some cases, the response is excessive due to persistence of the damage-causing agent (e.g., bacteria) or the offending trigger (e.g., immune complexes in autoimmune diseases). As will be discussed later, excessive or unregulated inflammatory responses can themselves cause tissue damage. In 1972, Lewis Thomas said "Our arsenals for fighting off bacteria are so powerful, and involve so many different defense mechanisms, that we are more in danger from them than the invaders. We live in the midst of explosive devices; we are mined." (Germs, *N Engl J Med*, 1972, 287:553–555.)

## Regulation of the Acute Inflammatory Response

How is the acute inflammatory response kept in check? It is clear that the response is subject to very tight regulation to contain the cascades before they lead to extensive tissue or organ injury. There are numerous, naturally occurring anti-inflammatory factors (Table 1.2). Cytokines such as IL-4, IL-10, and IL-12 in very low concentrations are inducible and are powerful anti-inflammatory factors that contain the acute inflammatory response by stabilizing IκBα, which blocks NF-κB activation. As a result, these regulatory cytokines have greatly diminished the production of proinflammatory mediators and reduced numbers of PMNs accumulating in tissues. There are several protease inhibitors that also contain the response by inhibiting serine proteases, many of which are released from phagocytic cells. The secreted leukocyte protease inhibitor (SLPI) was described as trypsin-like inhibitor that was largely confined to upper airway secretions, produced by and released from nearby epithelial cells of the lung. It is now known that SLPI reduces activation of NF-κB via stabilization of IκBβ. Hydrolysis of the IκB proteins is required for activation of NF-κB. In addition, there are nonserine protease inhibitors such as inhibitors of metalloproteases (MMPs). MMP3 and MMP9 may be the most important MMPs, with targets being elastin, collagen, and altered (denatured) collagens. Tissue inhibitor of MMP2 (TIMP-2) is a common and inducible TIMP and has broad inhibitory activity for MMPs. α1 Protease inhibitor ($\alpha_1$PI) is abundantly present in plasma and in lung tissue. It is a powerful serine protease inhibitor of trypsin-like enzymes. If $\alpha_1$PI is absent or present in a functionally defective manner, this is almost always associated with the development of progressive and often fatal pulmonary emphysema in humans. Such individuals may also develop hepatic cirrhosis for reasons that are poorly understood. There are many other naturally occurring protease inhibitors that suppress tissue damaging proteases associated with the induction of acute inflammatory response.

Antioxidant enzymes, such as superoxide dismutase, catalase, and glutathione peroxidase, are abundant in a variety of tissues and can be upregulated in the course of the inflammatory response such as those occurring after hyperoxia, bacterial infection, ischemia–reperfusion, and in various other situations. Upregulation of antioxidant enzymes is especially well documented in the lung, in the case of Gram-negative bacteria (e.g., *Escherichia coli*) lipopolysaccharide (LPS) can rapidly and powerfully upregulate these antioxidant enzymes. Superoxide dismutase converts superoxide anion ($\bullet O_2$) to $H_2O_2$, while catalase destroys $H_2O_2$, reducing it to water and molecular oxygen (Table 1.2). Glutathione peroxidase in the presence of glutathione (GSH) catalyzes the conversion of $H_2O_2$ to $H_2O$. If GSH levels are very low in an organ or tissue, it leads to "redox stress" in which the tissue has impaired ability to deal with oxidants

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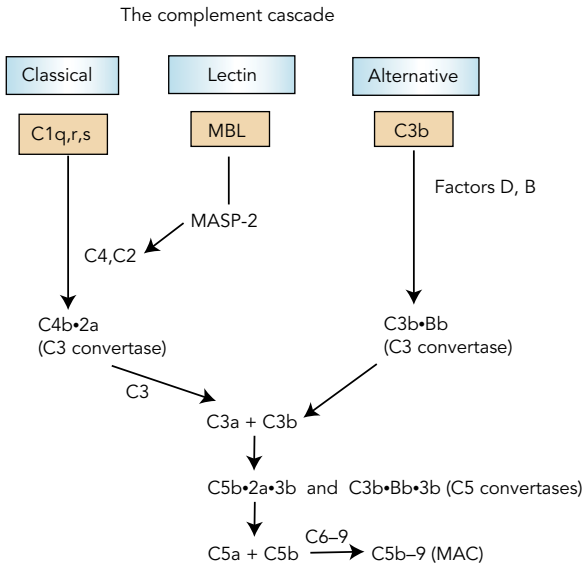



**Figure 1.7.** The complement cascade, including the three pathways of activation, the C3 and C5 convertases, and the chief complement activation products.

(Chapter 17). Lipoxins represent another source of natural anti-inflammatory factors (Chapter 12). Glucocorticoids are well known to be naturally occurring anti-inflammatory factors. Kininases hydrolyze kinins such as bradykinin and lysyl-bradykinin, leading to their functional inactivation. These enzymes are present in most tissues. The lung vasculature is lined with kininases, such that one pass of blood through the lung can result in complete inactivation of kinins. Phosphatases, such as PTEN, remove phosphates from proteins such as transcriptional and signaling factors, leading to their termination as functionally active moieties. Such regulation can greatly reduce the production of proinflammatory molecules (e.g., adhesion molecules, cytokines, chemokines, etc.). Similar to all other cascades of the inflammatory system, these phosphatases control the production of proinflammatory mediators. Finally, there are several transcriptional regulatory factors such as suppressor of cytokine signaling 3 (SOCS3) and STAT3 that block the activation of proinflammatory genes, resulting in greatly reduced levels of proinflammatory mediators. Obviously, all of these factors are aligned to bring about tightly regulated inflammatory responses before they unleash serious damage to tissues.

## THE COMPLEMENT CASCADE

The complement system is an important part of the innate immune system conferring protection especially against invading infectious agents, such as bacteria, viruses, and protozoa. Its role in innate immunity is to generate biologically active products from the pathways of complement activation, resulting in proinflammatory mediators that will call in PMNs as well as the production of opsonic (phagocytosis promoting) and lytic factors for bacteria (C5b-9, membrane attack complex [MAC]) and nucleated cells. The three pathways of complement activation are shown in Figure 1.7. The *classical pathway* is traditionally activated by the presence of IgG or IgM immune complexes. Activation of the first complement component (C1q,r,s) leads to activation of the subunits to active enzymes, with targets being C4 and C2, resulting in fragmentation products (C4a, C4b, C2a, C2b), some of which form the C4b•2a complex, which is a C3 convertase that cleaves C3 into C3a and C3b. C3b can be adducted to the C4b•2a complex to form the complex, C4b•2a•3b, which is a C5 convertase that can convert C5 into C5a and C5b. The second pathway of complement activation is the *lectin pathway* which involves the mannose-binding lectin (MBL), a plasma "collectin," that binds to mannose-related carbohydrates present on surfaces of viruses and bacteria. This leads to the activation of mannose-associated serine protease-2 (MASP-2), a serine protease that has the ability (like C1q,r,s) to interact with C4 and C2 to form the C3 convertase (C4b•2a). The third pathway of activation is the *alternative complement pathway* that can be activated by the presence of C3b which, when interacting with factors B and D, forms a complex, C3b•Bb, which has C3 convertase activity that generates C3a and C3b. Adduction of another molecule of C3b generates the C5 convertase of the alternative pathway, C3b•Bb•3b. The C5a convertases cleave C5 into C5a and C5b. C5b can interact with the terminal complement proteins, C6–9, to form the C5b–9 complex (MAC). In addition to these traditional pathways of complement activation, other serine proteases unrelated to the complement system can interact directly with C3 or C5 to form complement activation products (C3a, C3b, C5a, C5b). For instance, plasmin can interact with C3 to generate C3a and C3b. There are several serine proteases (such as the elastase present in neutrophils and a neutral protease present in macrophages) that will then interact directly with C5 to generate C5a and C5b. In addition, thrombin has the ability to interact with C5 to produce the same activation products. The complement activation pathways are under very rigid and tight control, based on "complement regulatory proteins" (CRPs) that are present both in plasma and on cell surfaces. These CRPs tightly regulate the complement system to either limit the formation of complement activation products or form a protective shield to prevent the activation products from bringing about cell damage. Some complement-mediated human disorders, such as paroxysmal nocturnal hemoglobinemia, result in intensive hemolysis of

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RBCs because of a defect in two of the CRPs (decay accelerating factor and CD59).

The complement anaphylatoxins are small peptides (<10 kDa) and consist of C3a, C4a, and C5a. The most abundant of these is C3a since C3 is the complement protein present in highest concentration in plasma. C3a appears to have its major biological activity as induction of histamine release from mast cells, which then leads to greatly increased vasopermeability in the local area. C3b is the major opsonic product generated by the complement system and reacts with receptors on a variety of different cells and microorganisms to bring about greatly enhanced phagocytosis and intracellular killing of microbes. There are relatively few humans with complete C3 deficiency and, as such, they are highly susceptible to life-threatening bacterial infections. The role of C4a is not well understood. C5a is an extremely potent anaphylatoxin which, in very low nanomolar concentrations, can interact with receptors on phagocytic cells, especially neutrophils, either to bring about their priming for enhanced subsequent responses in the presence of a co-stimulus or to bring about direct activation of phagocytic cells by inducing chemotaxis, an intracellular calcium response, generation of reactive oxygen species ($\bullet O_2$, $H_2O_2$), enzyme release, and a variety of other responses, all of which tend to function as a protective shield in a local setting and bring about accumulation of neutrophils at inflammatory sites. A major function is to contain and kill microorganisms. In some instances, excessive amounts of C5a are generated as in sepsis and in autoimmune diseases (such as rheumatoid arthritis and systemic lupus erythematosus [SLE]). In these cases, major problems can arise such as the signaling paralysis of neutrophils due to excessive generation of C5a and the priming of macrophages for accentuated and excessive inflammatory responses during sepsis. The final product of the complement activation sequence, C5b–9 (MAC), attaches to surfaces of antibody-coated bacteria, leading to their cytolytic destruction. In certain autoimmune disorders in which there are antibodies that can react with epitopes on surfaces of nucleated cells, cell lysis can occur. Soluble C5b–9 has the ability to interact with endothelial cells to bring about their activation with the formation of proinflammatory cytokines and chemokines. Finally, C5b–9 is an important protective factor leading to lysis of Gram-negative bacteria. Details on the biochemistry and functions of the complement system and its role in human diseases are discussed elsewhere.

### Intercommunications between Inflammatory Cascades and the Coagulation Cascade

Figure 1.7 demonstrates the intricate intercommunications between three different proinflammatory cascades: the kinin-generating cascade, the clotting cascades, and the fibrinolytic cascade. Central to these intercommunications is the clotting system which involves two activation pathways, the intrinsic and the extrinsic cascades. The intrinsic cascade occurs with the engagement and activation of Hageman factor (Factor XII) which interacts with Factors Va and VIIIa to convert Factor XII to XIIa ("a" signifies the active form of the protein). In turn, this leads to the activation of Factor X, which then directly converts prothrombin to thrombin. Thrombin converts fibrinogen to fibrin, which is the major product involved in in vivo clot formation. Following vascular injury, the extrinsic clotting cascade is activated resulting in the expression of endothelial cells on the surfaces of TF and in the copresence of other clot activating factors (Xa, IXa, VIIIa, Va), there is also conversion of prothrombin to thrombin and generation of fibrin from fibrinogen. The fibrinolytic cascade is activated by urinary plasminogen activator (uPA) and TPA which cause conversion of plasminogen to plasmin. Plasmin directly interacts with fibrin to bring about fibrin degradation products resulting in the breakdown of fibrin clots as they are formed within the intravascular compartment or elsewhere. TPA is used in patients with acute myocardial ischemia to try to bring about lysis of intracoronary arterial clots to allow perfusion to occur. The kinin-generating cascade is also linked to the clotting cascades by the fact that Factor XIIa will convert prekallikrein to kallikrein. Kallikrein interacts with high-molecular-weight kininogen (HMWK) to bring about its hydrolysis and release of bradykinin, which is a powerful vasopermeability agent. Bradykinin also has the ability to slow the heart rate (bradycardia). All of these cascades as well as the complement cascade have, as a common theme, activation of proteins by their limited hydrolysis, after which the split products directly interact with cell receptors to trigger cell responses (e.g., C5a interacting with receptors on PMNs [see earlier]) or the split products can assemble to form an active enzyme (such as C4b•2a, the substrate of which is C3). As with the complement system, the clotting, generating, and fibrolytic systems are each subject to very tight regulation by a series of inhibitors designed to prevent excessive product formation when one of the cascades is activated.

## OUTCOMES OF THE ACUTE INFLAMMATORY RESPONSE AND DISORDERED RESPONSES

A clinical example of an acute inflammatory response that is not adequately contained is the acute respiratory distress syndrome (ARDS) in humans where there is a sustained accumulation of PMNs within the distal airway (alveolar) compartment. ARDS occurs in adults and in infants in a variety of clinical

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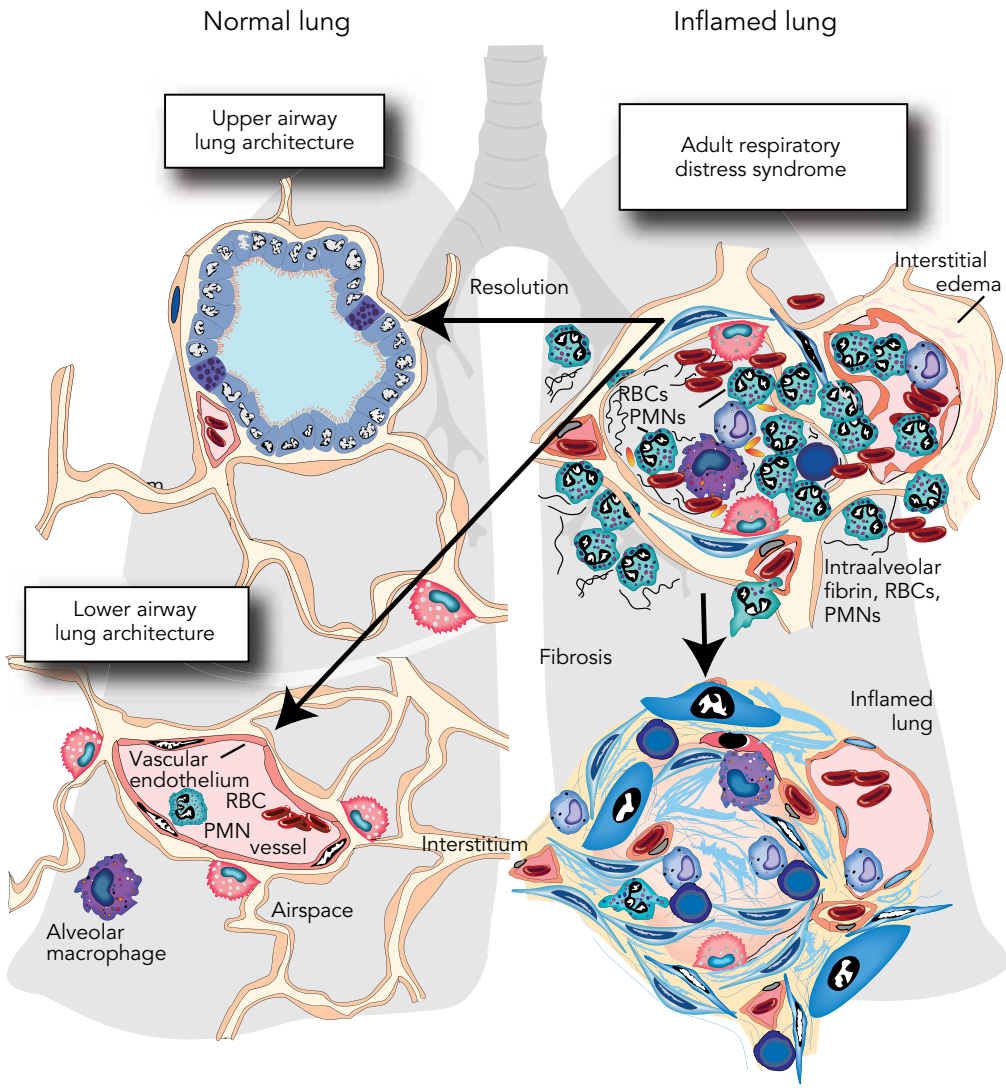



**Figure 1.8.** Outcomes of the acute inflammatory response in lung in ARDS in humans. In ARDS, edema fluid, fibrin, and PMNs accumulate in the alveolar compartment (upper right area). The inflammatory response can regress (undergo resolution) with return of the lung to its preinflammatory state (lower left area), or it may progress to extensive pulmonary fibrosis (lower right area).

situations, such as premature birth and polytrauma and bacterial pneumonia in adults. The mechanisms responsible for the development of ARDS are very poorly understood. ARDS can be considered to be a sustained and dangerous inflammatory condition in the lung. Bronchoalveolar fluids contain an abundance of PMNs, fibrin split products, and C5a. There is no known specific therapy for ARDS, only supportive treatment (mechanical ventilation, fluid therapy, etc.). ARDS may proceed to resolution or to pulmonary fibrosis that is often fatal (Figure 1.8). The pulmonary infiltrates in ARDS patients often lead to a radiographic "whiteout" in lungs, due to alveolar edema, PMNs accumulation, and fibrin deposition, collectively causing severely compromised gas exchange between the alveolar and vascular compartments, resulting in a high mortality rate. Why some cases of ARDS resolve completely (Figure 1.8) while others progress to pulmonary fibrosis and pulmonary arterial hypertension (Figure 1.8, lower right frame) is entirely obscure. Another clinical example in which the acute inflammatory response is not sufficiently contained is in the setting of sepsis, following bacterial or viral pneumonia, intestinal perforation, or any number of other clinical situations. Sepsis occurs in all age groups and is age related, commonly seen in patients who are immunocompromised (postchemotherapy or age related). The death rate can be as high as 60%. Over the past decade there has been a progressive rise in the incidence of sepsis, which has been linked with bacteria, both Gram-positive (such as *Staphylococcus aureus*) and Gram-negative (such as *E. coli*), viruses, and fungi. Sepsis is clearly a condition in which there has been loss in control of the inflammatory system. For instance, there is a surge of proinflammatory mediators (e.g., TNFα, IL-1β, IL-6,

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etc.) in the plasma, referred to as the "cytokine storm," although, the cause for high levels of these mediators is poorly understood. The vasculature undergoes upregulation of adhesion molecules for leukocytes. Tissue macrophages are "primed" and show excessive responses (e.g., production of oxygen free radicals, granule enzyme release, etc.). Blood PMNs express high levels of adhesion molecules such as CD11b/CD18 on their surfaces, which indicate PMN activation. In sepsis, there is extensive evidence of complement activation and consumptive coagulopathy (activation of the clotting system with depletion of the clotting factors) based on products measurable in plasma. Why the body has lost control of the inflammatory system during sepsis is entirely unknown. Currently, in addition to traditional supportive therapy (fluid resuscitation, ventilatory support, broad spectrum antibiotics, etc.), only a single drug (activated protein C, APC) has been approved for therapy in sepsis. Because APC is an anticoagulant and most septic patients have consumptive coagulopathy, APC is of limited value in sepsis.

## FUNCTIONAL CONSEQUENCES OF THE ACUTE INFLAMMATORY RESPONSE

Once the acute inflammatory response has been triggered in tissues, it is important to understand the consequences. Initially, the acute inflammatory response can be triggered by the presence of infectious agents such as bacteria. The rapid build-up of plasma constituents in the lung alveolar space results in accumulation of antibodies, complement proteins, clotting factors, and other factors that may assist in containment of microorganisms. Traumatic injury triggers an acute inflammatory response in a locale, although in spite of the localized nature of this response, there are often systemic symptoms such as fever, increased numbers of circulating neutrophils (neutrophilia), increased heart rate (tachycardia), and sometimes a feeling of anxiety and apprehension. The local consequences of traumatic injury are well known. The common situation of a sprained ankle results in increased blood flow to the local area which is characterized by redness and increased local temperature. Soon, extensive edema develops, causing soft tissue swelling and pain in the joint area which prevents a full range of motion. Induration (increased thickness of soft tissue) is due to the accumulation of leukocytes and edema fluid. Occasionally, hemorrhage may develop. As described earlier, the acute inflammatory response may resolve with ultimate removal of cell debris, fibrin, red cells, and disappearance of leukocytes. Well-known therapy consists of immobilization of the joint and application of localized cold temperature to reduce blood flow to the area. (The inflammatory response absolutely requires accelerated blood flow to the area.) Sometimes, the acute inflammatory response will not resolve for various reasons, such as the inability to clear an infectious agent whose contained presence will "fan" the flames of the inflammatory response, resulting in persistent and sometimes increasing intensity of inflammation. Ischemia–reperfusion of any organ or region will trigger an acute inflammatory response during the phase of reperfusion when sufficient blood flow is occurring to allow formation of edema and extravascular accumulation of leukocytes, primarily neutrophils. Toxic agents are well known to be able to cause tissue injury and unleash an inflammatory response. In the liver, excessive ingestion of acetaminophen, which is metabolized by liver into free radical intermediates, can result in hepatocellular toxicity (necrosis). If the dose of the ingested drug is limited, the liver can regenerate and replace the lost hepatocytes. Weeks or months later, little evidence of the cell destructive events after drug ingestion may be apparent if regeneration is successful. If the dose of

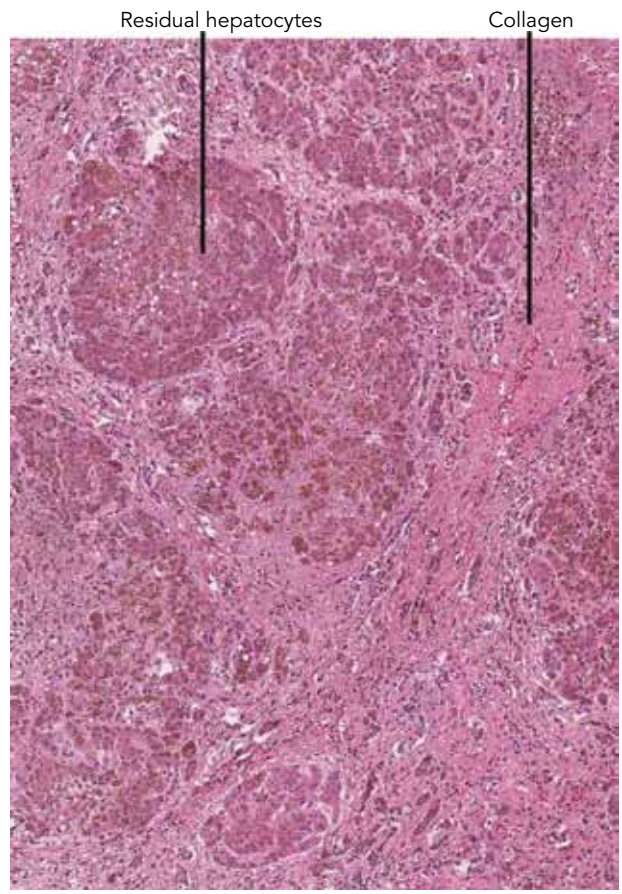



**Figure 1.9.** Postalcoholic cirrhotic liver. There are residual lobules of hepatocytes, with dense bands of collagenous scar surrounding some lobules and infiltrating others. The presence of dense collagenous scars interferes with the ability of residual hepatocytes to regenerate and restore damaged or destroyed hepatic lobules.

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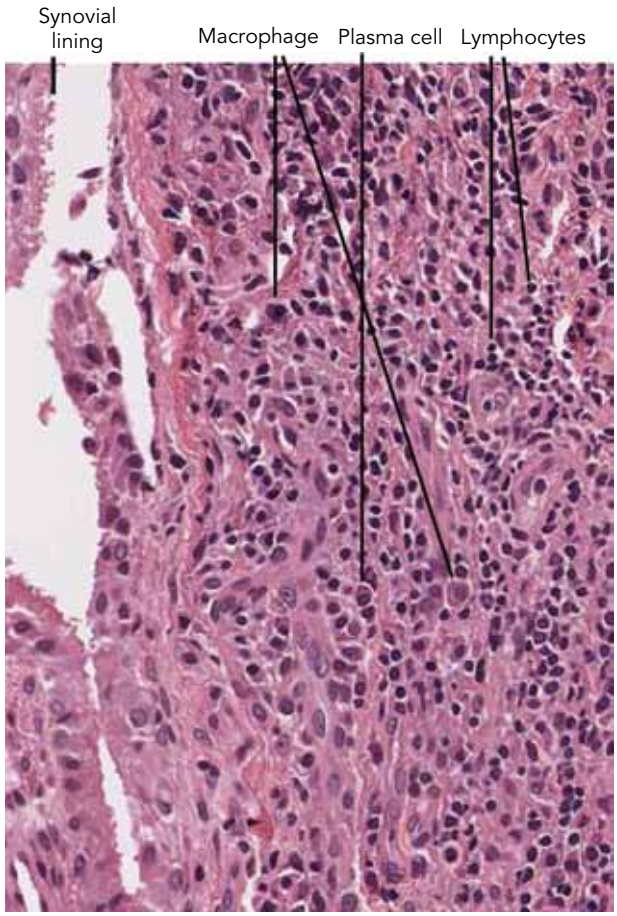



**Figure 1.10.** Synovium from knee joint of patient with rheumatoid arthritis. The synovium is thickened and contains lymphocytes, macrophages, and plasma cells.

ingested drug is higher, there may be too much damage for regeneration of the hepatic lobules, and bands of collagen may form around hepatic lobules along with chronic inflammatory cells, resulting in a condition referred to as cirrhosis (Figure 1.9). If the dose of ingested drug is very high, it may overwhelm the regenerative abilities of the liver, leading to death of the individual from acute hepatic failure. There are autoimmune diseases, such as rheumatoid arthritis or SLE, in which the predominant targets are joints where an acute inflammatory response persists together with the accumulation of acute and chronic inflammatory cells (Figure 1.10). This ultimately results in fibrosis of the synovial tissues and bone formation, causing fusion of articular bones, creating a "frozen joint." In the case of SLE, the kidney may be continuously bombarded with immune complexes and complement activation products, resulting in neutrophil accumulation, followed by ultimate scarring (hyalinization) of glomeruli and loss of filtration function (Figure 1.11). In other organs such as in the heart, the accumulation of inflammatory cells in the conducting system may lead to damage of cells, resulting in arrhythmias, which are sometimes fatal. Outcomes of the acute inflammatory response may be resolution (as described earlier) or persistence of the inflammatory response, sometimes leading to replacement of PMNs by chronic inflammatory cells (lymphocytes, macrophages).

### Granulomatosis Inflammation

Granuloma formation is a special type of inflammatory response, consisting of globular (granular) accumulations of inflammatory and fibrotic nodules, together with chronic inflammatory cells (lymphocytes, macrophages) and giant cells. In tuberculosis, the central part of these granulomas usually undergoes necrosis (caseous necrosis) so that the tissue is largely taken out of functional usefulness (Figure 1.12).

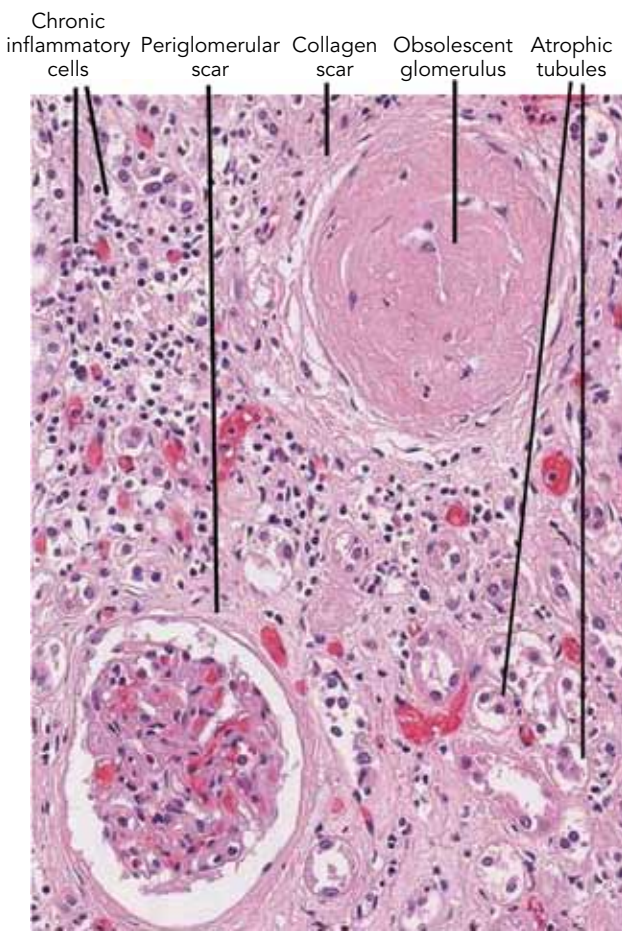



**Figure 1.11.** Outcome of persistent inflammation in kidney. One glomerulus is "obsolescent" (replaced with hyaline) and nonfunctional, contrasting to an intact glomerulus with thin periglomerular fibrous band (scar). In addition, there is interstitial fibrosis and the presence of chronic inflammatory cells. Extensive tubular atrophy has occurred, featuring greatly reduced numbers of tubular epithelial cells, and a greatly diminished mass of tubules.

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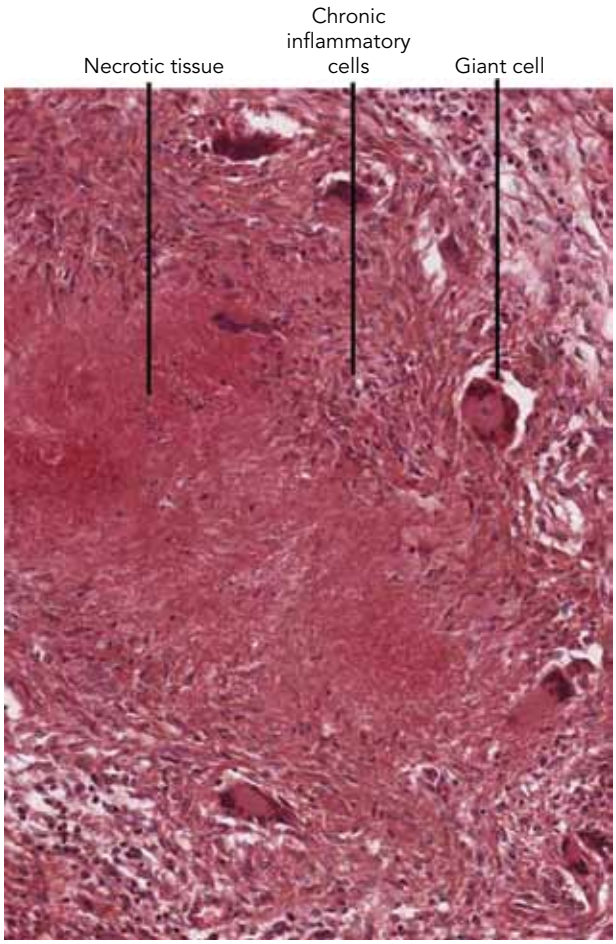



**Figure 1.12.** Lung mycobacterial granuloma featuring peripheral collagenous connective tissue along with multinucleate giant cells. In the center is necrotic (caseous) tissue, at the edge of which are macrophages and lymphocytes.

These types of inflammatory responses can result in extensive scarring of the affected tissue, with extensive collagenous scars and the development of fibrosis. In addition, tuberculosis granulomas, because of their necrotic centers, may ultimately erode into nearby pulmonary blood vessels, resulting in extensive pulmonary hemorrhage which is sometimes fatal. In granulomas that feature no central necrosis (and these would be the predominant type of granuloma in the United States where tuberculosis is rather infrequent), the causes may be fungi (e.g., histoplasmosis), foreign bodies (surgical suture material), or unknown causes (as in sarcoidosis). In sarcoidosis, there are granuloma nodules in tissues, especially in the lung, associated with the presence of fibroblasts and fibrotic tissue, chronic inflammatory cells (lymphocytes, macrophages, and giant cells) (Figure 1.13). Although necrosis is not seen in diseases like sarcoidosis, there may be extensive pulmonary fibrosis which ultimately leads to death because of interference with gas exchange (Figure 1.14). Under such conditions, there is often development of pulmonary hypertension. The combination of this complication, together with pulmonary fibrosis, may be fatal.

### Chronic Inflammation

Chronic inflammation is defined not as the persistence of acute inflammation but is defined morphologically by the presence of lymphocytes, macrophages, and plasma cells in tissues (Figure 1.15). In many cases, the chronic inflammatory response may persist for long periods (months to years). It is considered to be caused by persistent engagement of innate and acquired immune responses, such as in rheumatoid arthritis, in chronic allograft rejection, in beryllliosis, and in granulomatous inflammation, to list a few examples. There is evidence that macrophages in these lesions produce a series of proinflammatory mediators that activate fibroblasts to lay down collagen and activate other macrophages and lymphocytes

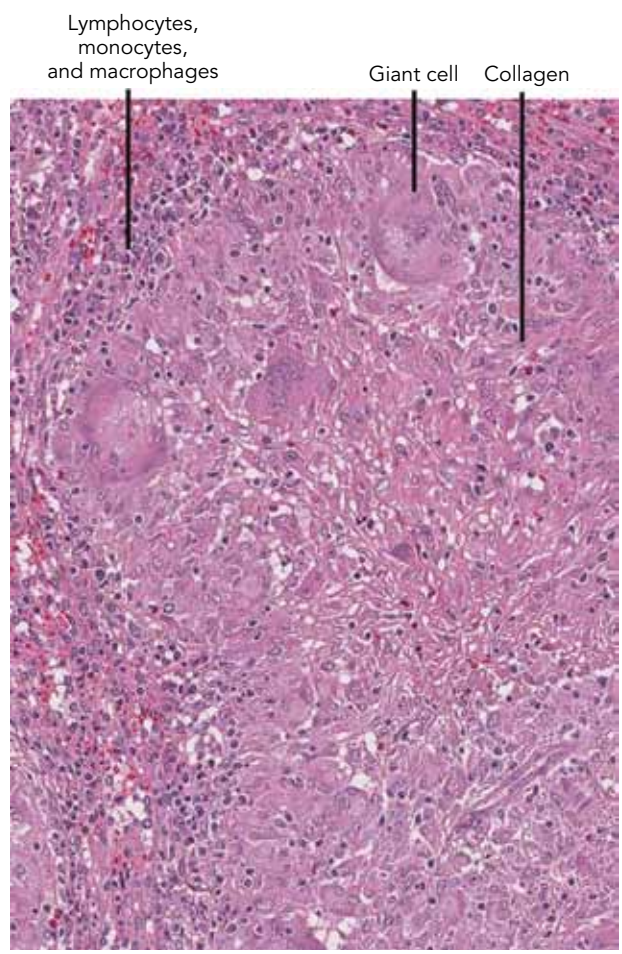



**Figure 1.13.** Lung biopsy of woman with sarcoidosis. There is a large granuloma surrounded by chronic inflammatory cells (lymphocytes, macrophages, and monocytes). The granuloma contains multinucleate giant cells, macrophages, and deposits of collagen. No necrosis is present.

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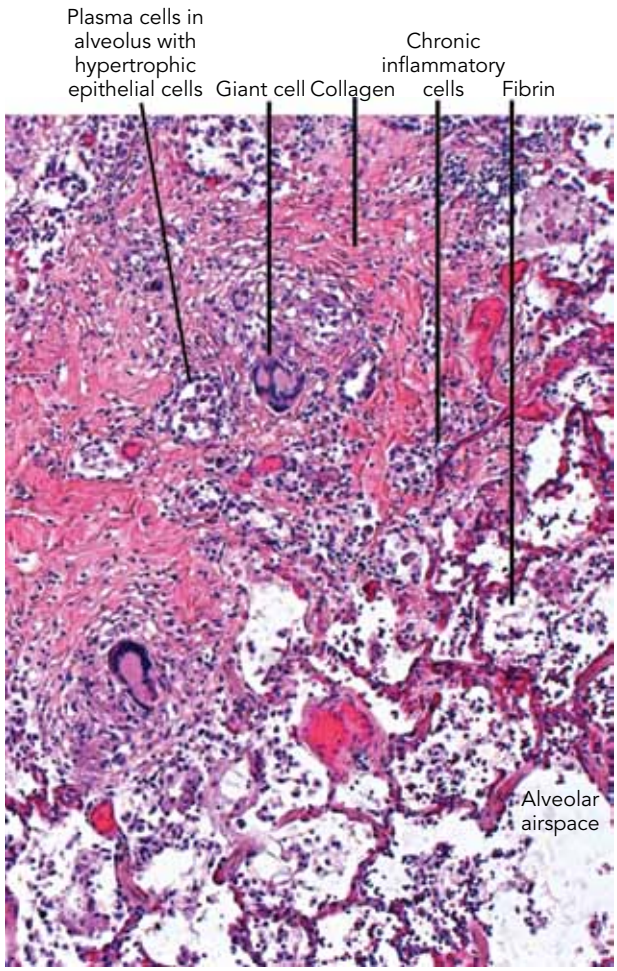



**Figure 1.14.** Pulmonary sarcoidosis. There are multinucleate giant cells, lymphocytes, and plasma cells replacing interstitial and alveolar spaces. Fibrin is also present. Dense collagenous deposits are starting to build-up in lung. No necrosis is present.

to release mediators to perpetuate these inflammatory responses. Why responses to mycobacterial agents result in extensive necrosis of the granulomas is unknown. As shown in Figure 1.16, chronic inflammation is initially triggered by vascular responses that involve the appearance of adhesion molecules on endothelial cell surfaces that will specifically cause adhesion of lymphocytes and mononuclear cells (monocytes), resulting in their subsequent transmigration into the extravascular compartment. In the case of lymphocytes, most are T cells. Like in the acute inflammatory response, lymphocytes and monocytes, as well as endothelial cells, undergo an activation process with lymphocytes and monocytes expressing adhesion molecules that are interactive with vascular (endothelial) adhesion molecules (such as VCAM-1 and other molecules) that promote adhesion and ultimate transmigration of these cells into the extravascular compartment. In any inflammatory response, differences between the types of adhesion molecules expressed on endothelial cells will determine the type of blood leukocytes (e.g., PMNs vs. monocytes vs. lymphocytes) that emigrate (Chapter 18). Since, in general, macrophages are not present in the peripheral blood, the influx of monocytes into the extravascular compartment allows these cells to differentiate into macrophages over a period of several days. In Peyer's patches in the small bowel, endothelial cells constitutively express adhesion molecules that specifically interact with T cells which then transmigrate into the lymphoid follicles (Chapter 11). Lymphocytes and macrophages in extravascular sites secrete factors (e.g., TGFβ) that will activate fibroblasts, resulting in the production of cross-linked collagen, sometimes resulting in extensive collagenous scars (Figure 1.17). In the case of plasma cells, the chief product is antibody which in the case of rheumatoid arthritis results in large amounts of IgGs being present both in the synovial tissue and in the synovial fluids bathing the

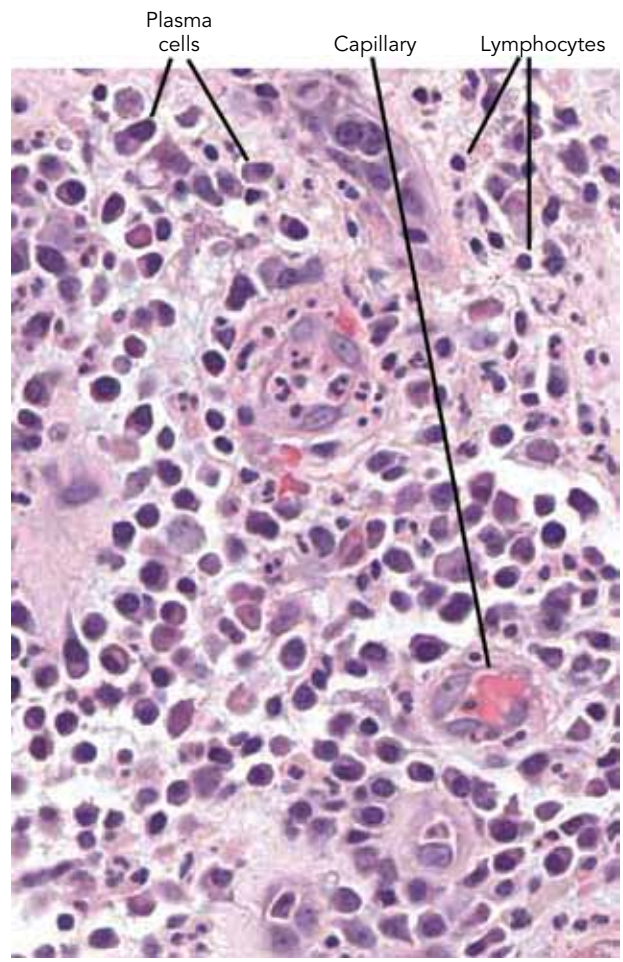



**Figure 1.15.** Chronic inflammation in soft tissue, featuring large numbers of plasma cells and chronic inflammatory cells (lymphocytes, mononuclear cells, and macrophages) in the interstitium.

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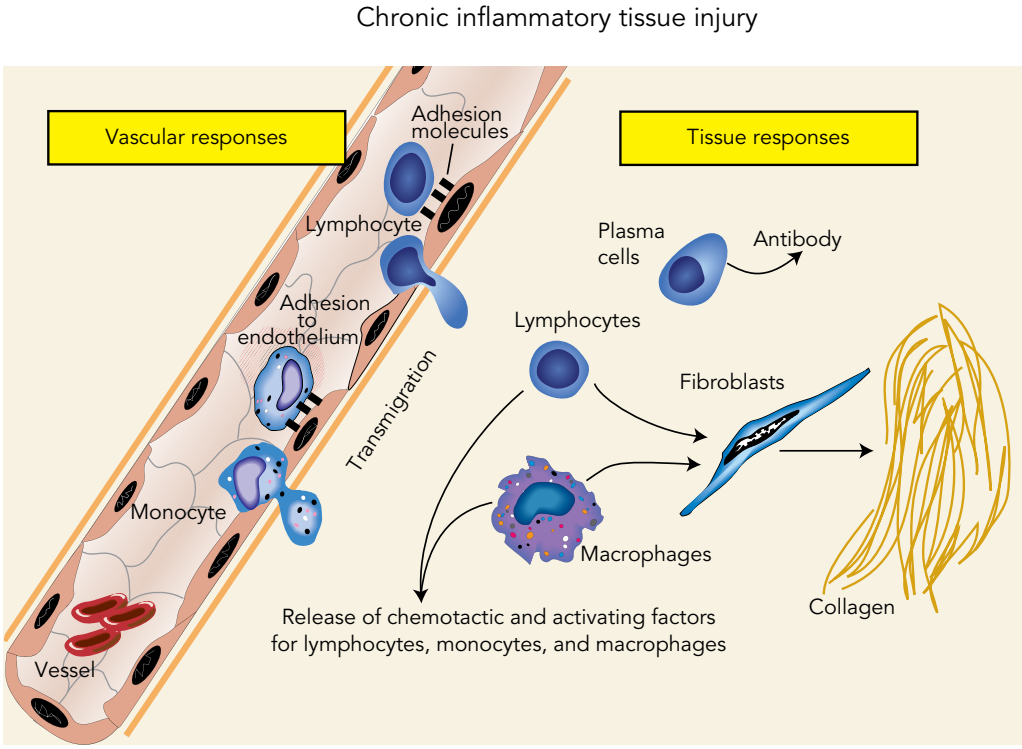



**Figure 1.16.** Mechanisms of chronic inflammation, with adhesion of lymphocytes and monocytes to the activated endothelium, and the eventual transmigration of these cells into the extravascular space. Activated endothelial cells express adhesion molecules (such as VCAM-1) that facilitate adhesion of lymphocytes and monocytes to endothelial surfaces, followed by their eventual transmigration. In the extracellular compartment, lymphocytes and macrophages secrete factors that stimulate extracellular collagen formation and perpetuate the inflammatory response. Transmigrated monocytes "mature" into macrophages. Plasma cells secrete various subclasses of antibodies.

joint space. The various factors produced by cells that are associated with collagen scar formation are discussed elsewhere (Chapter 13). With reference to chronic inflammation, this response may be short term or long term such as in the response to infectious agents especially such as mycobacterial species, protozoa, and so on. Chronic inflammation may resolve or there may be persistence of the chronic inflammatory response, as described later (Table 1.3). In persistent allograft rejection, recipient T cells infiltrate the transplanted organ and respond to donor histocompatibility antigens, releasing factors (cytokines and chemokines) that damage or even destroy the allograft (Figure 1.18). The purpose of immunosuppressive drugs is to prevent the build-up of T cells in the allografts, thereby preventing loss of function of the allograft or its destruction. There is a special type of rapid graft rejection (hyperacute rejection) which is usually associated with the presence of preformed antibody to graft antigens (often following prior blood transfusions received by the recipient whose immune system generates antibodies to antigens present on lymphocytes of the donor). In hyperacute allograft rejection, the process often starts within hours of vascular anastomosis of the graft and is characterized by endothelial damage in arterioles and capillaries, together with intravascular fibrin clots and PMNs. Immediate immunosuppressive and corticosteroid therapy is initiated. The result of persistent chronic inflammation, as seen in the autoimmune disease, scleroderma, can lead to an intense fibrotic response in organs such as the lung (Figure 1.19) or, as described earlier in the case of a heart allograft (Figure 1.18) to the destruction of cardiac tissue.

## THERAPEUTIC INTERVENTIONS IN INFLAMMATION

There are many situations when it is desirable to suppress the inflammatory response if the response is causing an immediate threat (such as abruptly rising intracranial pressure in a patient with bacterial meningitis) or represents a long-term threat (as in rejection of an allotransplanted organ). There are several principles

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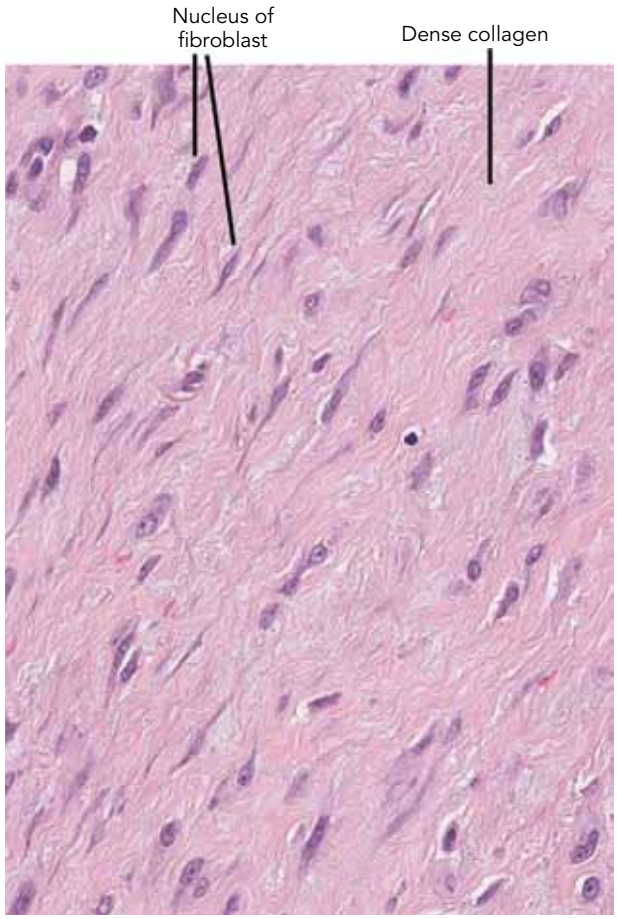



**Figure 1.17.** Dense collagenous scar with fibroblasts with little, if any, evidence of inflammatory cells. Cylindrical nuclei are in fibroblasts.

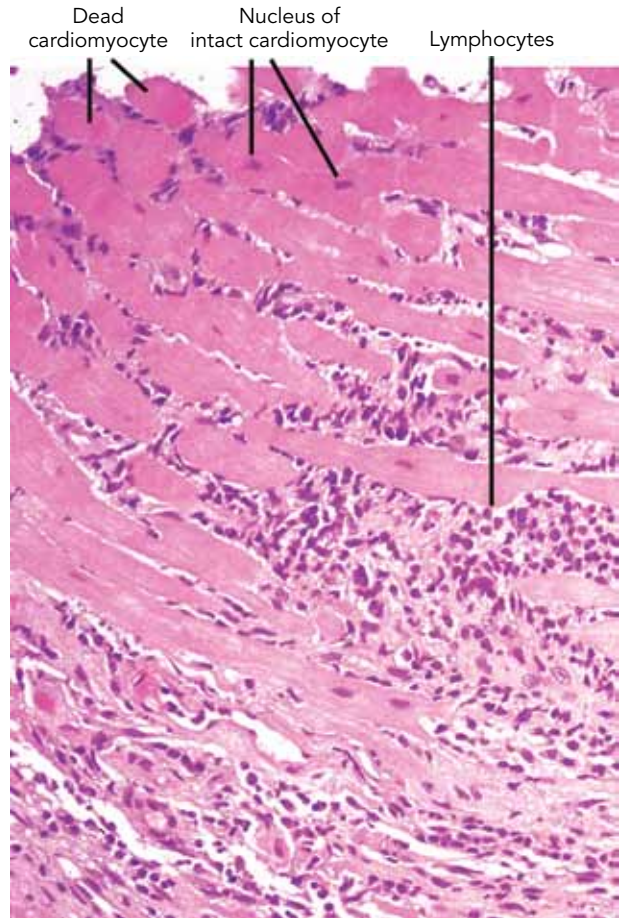



**Figure 1.18.** Features of chronic rejection of an allotransplanted human heart. Some cardiomyocytes are necrotic as manifested by loss of striations and nuclei and intensified eosin (red) staining. There is an intense accumulation of interstitial lymphocytes, which are known to be of recipient origin and involved in immunological damage (rejection) of the transplanted heart.

**TABLE 1.3. Consequences of acute inflammation**

| Responses | Outcomes |
|---|---|
| Acute inflammation | • Responses to infectious agents and their containment (e.g., bacteria)<br>• Response to trauma<br>• Response to persistence of trigger (e.g., infectious agents), to toxic agents, or to immunological responses (autoimmune, etc.)<br>• Resolution of persistence of inflammation or progression to chronic inflammation<br>• Outcome may be resolution (clearance of edema fluid, fibrin, RBCs, and leukocytes) – persistence of inflammation, sometimes leading to fibrosis |
| Chronic inflammation | • Response to persisting inflammatory trigger (infectious agents, autoimmune products, immune complexes)<br>• Inflammatory cells (lymphocytes, macrophages) may persist or be cleared. Giant cells may develop, together with fibrosis (as in sarcoid) and/or necrosis (as in mycobacterial infections) |

that need to be considered. The first is the cause of the inflammatory response. In many situations this cannot be clearly determined, as in rheumatoid arthritis and in certain types of glomerulonephritis. If the cause of the inflammatory trigger is known (as in meningitis or pneumonia caused by bacteria), specific therapy (antibiotic drugs) are immediately instituted. If the inflammatory reaction is causing serious functional problems (defective gas exchange in bacterial pneumonia), supportive therapy (mechanical ventilation,

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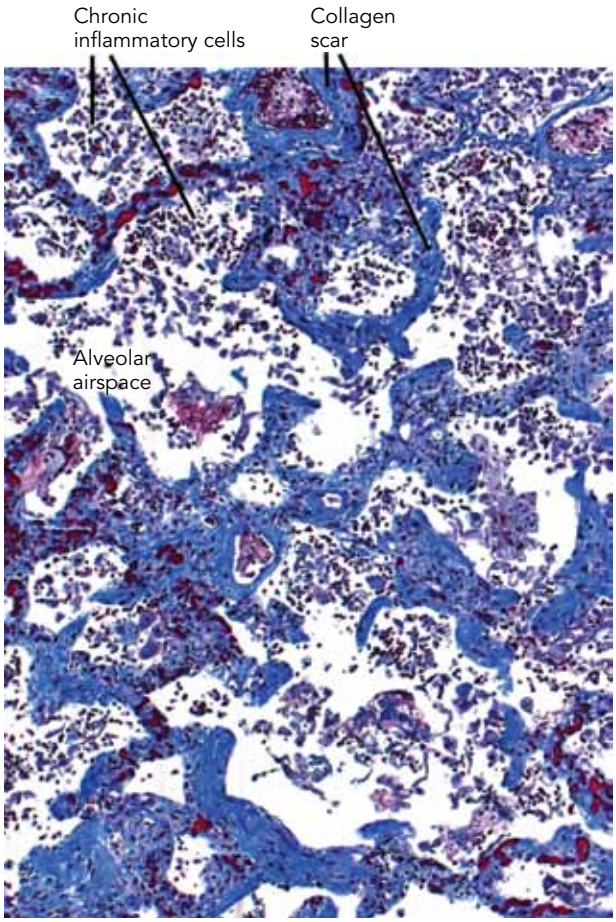



**Figure 1.19.** Scleroderma lung featuring extensive interstitial fibrosis (blue staining collagen) and alveolar spaces containing lymphocytes and fibrin. (Masson trichrome stain.)

high levels of oxygen, etc.) is instituted. Other interventions can also be employed. Corticosteroid therapy has powerful suppressive effects on all aspects of the inflammatory response. Nonsteroidal drugs (inhibitors of cyclooxygenases I and II) will provide symptomatic relief of pain but these drugs, on balance, have very limited effects on inflammation. That is, they poorly suppress the inflammatory response. If there is evidence that the inflammatory response is being driven by an immune response (as in autoimmune diseases such as SLE or rheumatoid arthritis, or in graft rejection or in many forms of glomerulonephritis), then immunosuppressive drugs are employed and the doses "titrated" up or down depending on clinical symptoms. In some inflammatory diseases, blockade of a single proinflammatory mediator (TNFα) has been found to be effective (rheumatoid arthritis, severe psoriasis, inflammatory bowel disease) using blocking antibody to TNFα or using the soluble TNFα receptor that intercepts TNFα with high affinity before TNFα can react with cell surface receptors to TNFα. Clearly, much more needs to be understood about the inflammatory response before more effective inflammation-blocking strategies are available and ones that, unlike corticosteroids, can be used over the long term without serious side effects.

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# 2 Resolution of Acute Inflammation and Wound Healing

*Derek W. Gilroy*


## SUMMARY

It is without doubt that resolution of acute inflammation is under strict checkpoint control by endogenous proresolution factors. It is these factors and mechanisms inherent in resolution that are crucial in preventing excessive tissue injury, autoimmunity, and chronic inflammation. In this chapter, resolution and the factors that control it are detailed to underline its importance in human pathology and highlight new and more effective treatment modalities with fewer side effects for chronic inflammatory diseases.


## INFLAMMATION IN HEALTH AND DISEASE

Inflammation is a beneficial host response to foreign challenge or tissue injury that leads ultimately to the restoration of tissue structure and function. It is a reaction of the microcirculation that is characterized by the movement of serum proteins and leukocytes from the blood to the extravascular tissue. This movement is regulated by the sequential release of vasoactive and chemotactic mediators, which contribute to the cardinal signs of inflammation – heat, redness, swelling, pain, and loss of tissue function (Figure 2.1A). Local vasodilation increases regional blood flow to the inflamed area and, together with an increase in microvascular permeability, results in the loss of fluid and plasma proteins into the tissues. Concomitantly, there is an upregulation of adhesion molecule expression on endothelial cells and the release of chemotactic factors from the inflamed site, which facilitate the adherence of circulating cells to the vascular endothelium and their migration into the affected area. These tightly regulated events result in a predominance of polymorphonuclear leukocytes (PMNs, see Glossary) in the inflamed area at the onset of the lesion, which are later gradually replaced by mononuclear cells – mainly monocytes, which then differentiate into macrophages. These phagocytic cells ingest foreign material and cell debris. They also release hydrolytic and proteolytic enzymes, and generate reactive oxygen species that eliminate and digest invading organisms. Finally, the injurious stimulus is cleared and normal tissue structure and function is restored [1].

However, inflammation can cease to be a beneficial event and contribute to the pathogenesis of many disease states. The chronic inflammatory disease rheumatoid arthritis, for instance, is characterized by the accumulation and persistence of inflammatory cells in synovial joints, which results in joint damage. This loss of tissue or organ function as a result of an inappropriate inflammatory response is also seen in various other diseases, such as chronic bronchitis, emphysema, asthma, glomerulonephritis, myocardial infarction, and ischemia reperfusion injury. By contrast, certain inflammatory diseases have an intrinsic capacity for complete resolution without tissue injury – for example, lobar streptococcal pneumonia, which involves the extensive accumulation of PMNs, monocytes, and macrophages in the lungs. Studies of patients who have lobar pneumonia show that most of the lesions resolve without any evident tissue destruction. Experiments in animal models of streptococcal pneumonia show resolution of tissue pathology within days. Therefore, this type of self-limiting inflammatory response is under the strict control of endogenous mechanisms. As continual activation of the adaptive immune system is the driving force behind chronic inflammation, it is crucial to identify the stop signals that are present in self-limiting, self-resolving inflammatory lesions. These signals might be used therapeutically to control the activation of the adaptive immune response and the transition from acute to chronic inflammation, when these signals might be absent or become dysregulated.

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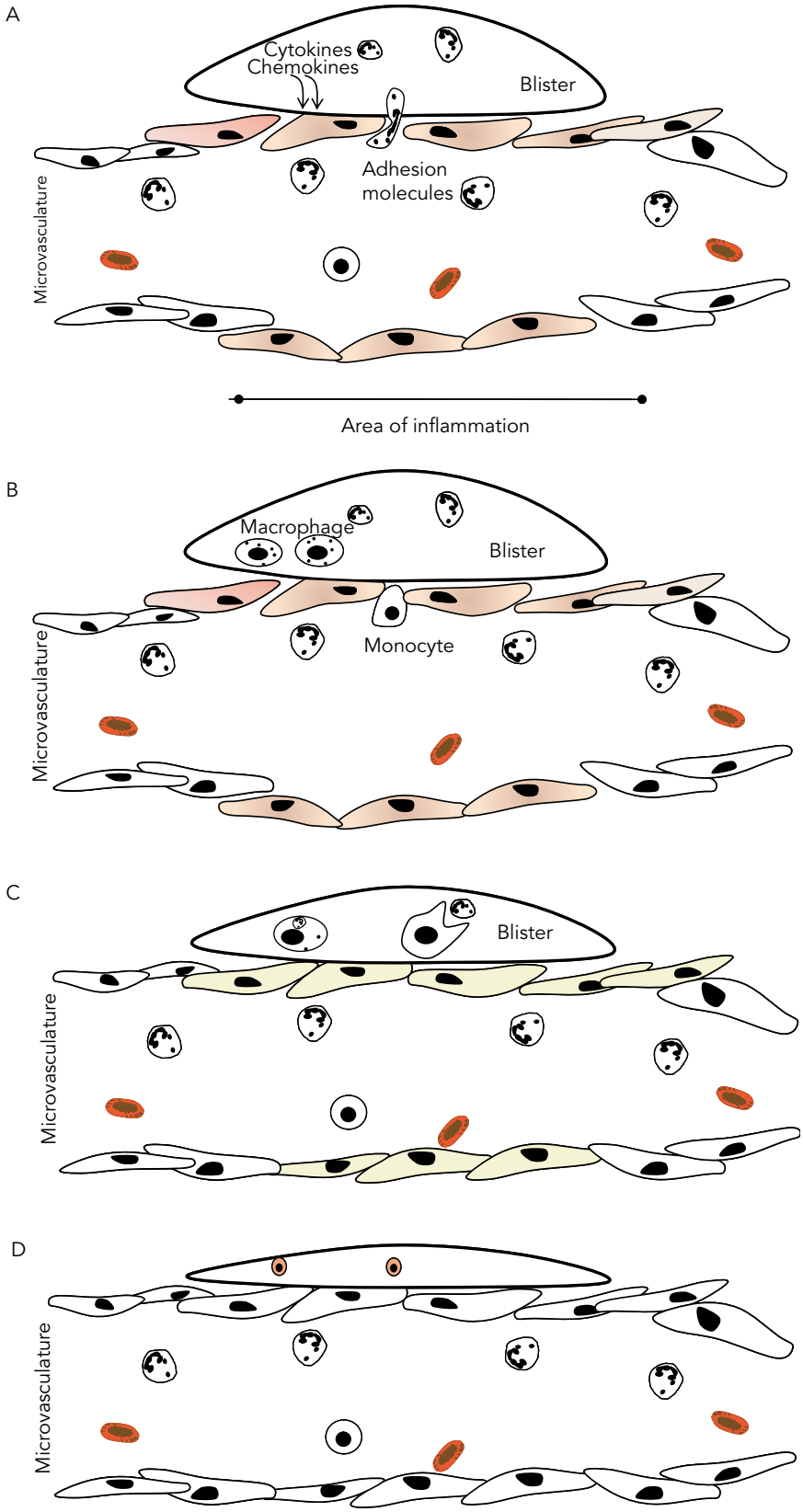



**Figure 2.1.** Simple depiction of acute inflammation and its resolution in the form of a skin blister. Depending on the nature of the injurious stimuli, the fist sequence of events (A) is communication between the site of injury and the microvasculature that serves the area, whereupon (B) endothelial cells become activated and recruit PMNs. In turn and provided there is no infection, whereupon PMNs will ingest and kill bugs, (C) monocyte migrate from the microvasculature to the extravascular space, differentiate into macrophages, which have a massive capacity for phagocytosing dead and apoptotic leukocytes. (D) In time may either die locally or drain via the lymphatics. Inflammation wanes and homeostasis is restored.

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Therefore, in this chapter we will discuss one of the many aspects of the inflammatory response – how acute inflammation resolves. We will also discuss wound healing and the type of resolution associated with substantive tissue injury. In doing so it will be argued that resolution is an active process, whose failure may predispose the host to chronic inflammatory diseases and autoimmunity such as that typified by rheumatoid arthritis and asthma. At the very least it is hoped that this chapter will highlight resolution as a critical facet of the inflammatory response and serve to underline the importance of not altering its normal course of action when developing novel anti-inflammatory drugs. Ultimately, it will be proposed here that resolution is controlled by endogenous pro-resolution factors, which, for the future, may represent a treasure trove for drug discovery in terms of designing drugs that mimic their mode of action or enhance their synthesis [2–4].

## WHAT IS INFLAMMATORY RESOLUTION?

To define the fundamental requirements for the successful resolution of either acute innate or acute adaptive immunity, it is becoming increasingly clear that the most simple but absolutely critical determinant for the inflammatory response to switch off is the neutralization and elimination of the injurious agents that initiated it in the first place. Failure to achieve this first step will invariably lead to chronic inflammation with the nature of the agent in question almost certainly dictating the etiology of the developing chronic immune response. For instance, chronic granulomatous disease is characterized by severe, protracted, and often fatal infection, which results from a failure of the phagocytic NADPH oxidase enzyme system to produce superoxide and kill invading infections leading to a predisposition to recurrent bacterial and fungal infections and the development of inflammatory granulomas [5]. Successfully dispensing with the inciting stimulus will signal a cessation to proinflammatory mediator synthesis (eicosanoids, chemokines, cytokines, cell adhesion molecules, etc.) and lead to their catabolism (Figure 2.1B). This would halt further leukocyte recruitment and edema formation. These are probably the very earliest determinants for the resolution of acute inflammation, the outcome of which signals the next stage of cell clearance. The clearance phase of resolution, be it innate PMN or eosinophil driven or adaptive (lymphocyte mediated), also has a number of mutually dependent steps. The clearance routes available to inflammatory leukocytes include systemic recirculation or local death of influxed PMNs, eosinophils, or lymphocytes followed by their phagocytosis by recruited monocyte-derived macrophages (Figure 2.1C). Once phagocytosis is complete, macrophages can leave the inflamed site by lymphatic drainage with evidence that a small population may die locally by apoptosis. If all of these pathways are strictly followed then acute inflammation will resolve without causing excessive tissue damage and give little opportunity for the development of chronic, nonresolving inflammation (Figure 2.1D). In the following, we will discuss the cellular changes that occur throughout acute inflammation that ultimately leads to resolution and the soluble mediators that temper the severity of onset as well as trigger resolution. Finally, we will differentiate between resolution of acute inflammation (leukocyte clearance and inflammatory mediator catabolism) and wound healing, a form of inflammatory resolution associated with parenchymal tissue injury.

## SOLUBLE MEDIATORS OF INFLAMMATION AND RESOLUTION

### Controlling Onset

It is well known that inflamed tissues generate local proinflammatory stimuli to drive acute inflammation including cytokines, chemokines, and cell adhesion molecules. Indeed, these have acted as targets for drug development in the form of anti-TNFα inhibitors (etanercept, infliximab), for instance. The original targets for the treatment of inflammation-driven disease were the cyclooxygenase (COX)-derived prostaglandins (PGs), with nonsteroidal anti-inflammatory drugs (aspirin, indomethacin, ibuprofen, naproxen) serving as the clinical inhibitor [6,7]. Interestingly, however, the PGs and the arachidonic acid metabolic cascade (COX, lipoxygenase [LOX] pathways) are now recognized to possess potent anti-inflammatory and proresolving properties (see later). Along these lines, there is a range of other systemic and locally produced endogenous mediators that counterbalance the severity of inflammatory onset. Studies in the 1950s and 1960s identified endogenous anti-inflammatory mediators that counteract vascular leakage – namely, adrenaline, noradrenaline, and 5-hydroxytryptamine – and intracellular cyclic AMP, a second messenger induced by several hormones; inflammatory mediators; and cytokines, which dampens leukocyte activation. Elevation of the level of intracellular cAMP – by inhibiting the enzyme system that is responsible for its catabolism (phosphodiesterase) – ameliorates immune and nonimmune inflammation in vivo and suppresses various cellular processes in vitro, including the immunological release of histamine and leukotrienes from mast cells, monocytes, and PMNs; lysosomal enzymes and reactive oxygen species from PMNs; and cytokines and nitric oxide (NO) from macrophages [2]. These data further

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indicate that cAMP has a central role in the resolution of inflammation. Perhaps the most powerful endogenous anti-inflammatory agents to be described so far are the glucocorticoids [8]. Glucocorticoids and their synthetic mimetics are used for the treatment of several chronic inflammatory diseases, including rheumatoid arthritis, inflammatory bowel disease, asthma, psoriasis, and vasculitis. Most of the actions of glucocorticoids require binding to cytoplasmic steroid hormone receptors that migrate to the nucleus and antagonize proinflammatory gene transcription. However, glucocorticoids also induce the expression of regulatory proteins that have anti-inflammatory actions, of which the peptide annexin 1 (previously known as lipocortin 1) has been well described in vitro and in vivo [9]. Annexin 1 has been shown to inhibit the production of PGs, as well as PMN and monocyte migration, in vivo. The take home point here is that although there are signals that drive the onset of inflammation (PMN trafficking) there are also endogenous factors released by stromal and/or hematopoietic cells that temper the severity of this response, save inflammation becomes too severe or inappropriate in magnitude.

## Onset to Resolution

One well-described event in the transition toward resolution is the replacement of PMNs or eosinophils by monocytes and phagocytosing macrophages. However, until recently our understanding of the signals that control this cell profile switch was unclear. Studies addressing this issue of leukocyte infiltration in peritoneal inflammation have suggested that the interaction between interleukin (IL)-6 and its soluble receptor, sIL-6R, forms one of the major determinants of this switch from PMNs to monocytes. It was shown that sIL-6R, produced by the infiltrating PMNs, forms a complex with IL-6 which, in turn, directly modulates CC and CXC chemokine expression. Thus, CXC chemokine synthesis, induced by IL-1 and tumor necrosis factor (TNF)α, was suppressed whereas the CC chemokine monocyte chemoattractant 1 was promoted. This chemokine shift suppresses further PMN recruitment in favor of sustained mononuclear cell influx. In addition to chemokines, the eicosanoids also orchestrate the early transition to resolution in acute inflammation. Transcellular metabolism of arachidonic acid by LOX/LOX interaction pathways gives rise to the lipoxin (LX) family of eicosanoid metabolites. LXs display selective actions on leukocytes that include inhibition of PMN chemotaxis, PMN adhesion to and transmigration through endothelial cells, as well as PMN-mediated increases in vascular leakage (see Glossary). It is unclear at this point whether there is any cross talk between the LXs and IL-6/sIL-6R complex signaling in the control of leukocyte profile switching. Nonetheless, it seems that when acute inflammation needs to resolve the IL-6/sIL-6R, chemokines, and LXs representing some of the earliest signals that control the switch from very early PMNs to monocyte/macrophage. Over and above this, there is the need for proinflammatory mediator catabolism, specifically the removal of cytokines and chemokines that drive inflammation. To do this, D6, a scavenger receptor expressed on lymphatic endothelial cells binds and neutralizes inflammatory members of the b-chemokine family but not constitutive b-chemokines or members of the other chemokine subfamilies such that its absence predisposes to failed resolution. Indeed, CCR5 expression on apoptotic PMNs and apoptotic T cells also sequester and effectively clear CCL3 and CCL5 from sites of inflammation, with scavenging CCR5 expression being inhibited by proinflammatory TNFα, for instance, but upregulated by proresolving lipids (LXs and resolvins). Thus, there is the coordinated interaction with factors that drive inflammation (PMN influx, cytokines, chemokines, vasoactive amines) counterbalanced by endogenous factors that temper the severity of inflammation (PGD2, LXs, cAMP) overshadowed by factors that ensure the smooth transition to resolution.

## Resolving Inflammation

Studies on the resolution of acute inflammation have already revealed novel mediators with potent anti-inflammatory properties. Determining their basic structure and function might help in the development of unique anti-inflammatory therapeutics. So far, these proresolving mediators have been shown to exert powerful anti-inflammatory effects in various experimental models of inflammatory diseases. Of these, lipid-mediator derivatives of the COX-2 and LOX/LOX interaction pathways of arachidonic acid, eicosapentaenoic acid (EPA), and docosahexaenoic acid (DHA) metabolism will be described [10].

$PGD_2$ has emerged recently as an eicosanoid with both pro- and anti-inflammatory properties. $PGD_2$ undergoes dehydration in vivo and in vitro to yield biologically active PGs of the $J_2$ series, including $PGJ_2$, $\Delta^{12,14}$-$PGJ_2$, and 15-deoxy-$\Delta^{12,14}$-$PGJ_2$ (15d-$PGJ_2$). In addition to being a high-affinity natural ligand for anti-inflammatory peroxisome proliferator-activated receptor gamma (PPARγ) (PPAR is explained in the Glossary), 15d-$PGJ_2$ also exerts its effects through PPARγ-dependent and -independent mechanisms to suppress proinflammatory signaling pathways and the expression of genes that drive the inflammatory response. 15d-$PGJ_2$ also preferentially inhibits monocyte rather than PMN trafficking through the differential regulation of cell adhesion molecule and

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chemokine expression. We have shown that COX-2–derived $PGD_2$ metabolites contribute to the resolution of acute inflammation (pleuritis) through the preferential synthesis of $PGD_2$ and 15d-$PGJ_2$, which, along with the alternative DNA-binding p50–p50 homodimers complexes of nuclear factor kappa B (NF-κB) (see Glossary), bring about resolution by inducing leukocyte apoptosis. Indeed, there is an increasing body of evidence detailing the differential effects of $PGD_2$ metabolites on leukocyte apoptosis as well as the signaling pathways involved. In addition to the well-known eicosanoids, there is a new generation of lipid mediators showing promise as endogenous anti-inflammatories. Resolvins and docosatrienes are fatty-acid metabolites of the COX/LOX pathways, where the omega-3 fatty-acid constituents of fish oils (DHA and EHA) are the substrates and not arachidonic acid. Thus, transcellular metabolism of arachidonic acid by LOX/LOX interaction pathways gives rise to the LX family of eicosanoid metabolites. LXs display selective actions on leukocytes that include inhibition of PMN chemotaxis, PMN adhesion to and transmigration through endothelial cells, as well as PMN-mediated increases in vascular permeability. In contrast to their effects on PMN and eosinophils, LXs are potent stimuli for peripheral blood monocytes, stimulating monocyte chemotaxis and adherence without causing degranulation or release of reactive oxygen species. In fact, LXs and their stable analogues accelerate the resolution of allergic pleural edema and enhance phagocytosis of apoptotic PMNs by monocyte-derived macrophages in a nonphlogistic fashion (see Glossary), paving the way for a return to tissue normality. LXA4 and aspirin-triggered 15-epi-LXA4, as well as their stable analogues, act with high affinity at a G-protein–coupled receptor, LXA4 receptor (ALXR; also referred to as formyl peptide receptor-like 1 or FPRL1). FPRL1 is a member of the family of seven transmembrane G-coupled receptors, which has at least two other members – FPRL2 and the formyl-Met-Leu-Phe receptor (FPR). By contrast, LXB4 does not bind the ALXR and, although functional studies have indicated the existence of a receptor that is activated by LXB4, this receptor has not been cloned. As with the cyPGs, the LXs have also been identified as being expressed during and being crucially important for the resolution of acute inflammation. In a model of rat allergic edema, for instance, LXA4 was identified along with $PGE_2$ as being present during the clearance of edema in this model. Inhibition of their synthesis prolonged edema clearance, which was rescued using stable analogues of these eicosanoids. A recent analysis of eicosanoid synthesis in a murine dorsal air pouch of acute inflammation elicited by TNFα has revealed a switch in lipid class metabolism reminiscent of that found in the rat carrageenin-induced pleurisy. In response to TNFα, levels of leukotriene B4 increased rapidly, followed by PMN infiltration, which coincided with a rise in inflammatory exudate $PGE_2$. Concomitant with the eventual reduction in PMN numbers and $PGE_2$ was an increase in LXA4. It was concluded that $PGE_2$ induced a switch in lipid mediator synthesis from predominantly 5-LOX-generated leukotriene B4 to 15-LOX-elicited proresolving LXA4. Along with our findings in the rat carrageenin-induced pleurisy in terms of PG metabolism, this work indicates that, in acute inflammation, lipid-mediator biosynthesis is biphasic, with a role for eicosanoids in the initiation as well as termination of the inflammatory response [2].

Arachidonic acid is not the only fatty-acid substrate that can be transformed by COXs and LOXs to bioactive mediators with roles in anti-inflammation and resolution. DHA and EPA – omega-3 fatty-acid constituents of fish oils – were shown recently to be metabolized during the resolving phase of an aspirin-treated TNFα-induced inflammation to potent anti-inflammatory products, named resolvins. For instance, endothelial cells expressing COX-2 and treated with aspirin convert EPA to 18*R*-hydroxyeicosapentaenoic acid (HEPE) and 15*R*-HEPE. Both are subsequently used by PMNs to generate separate classes of novel trihydroxy-containing mediators that potently inhibit human PMN transendothelial migration. Similarly, aspirin-acetylated COX-2 converts DHA to 17*R*-HDHA, which is subsequently transformed by PMNs into two sets of novel di- and trihydroxy products that can inhibit microglial cell cytokine expression and ameliorate experimental models of dermal inflammation and leukocyte accumulation in peritonitis at nanogram doses. Even in the absence of aspirin, human whole blood converts DHA to 17*S* series resolvins as well as novel dihydroxy-containing docosanoids. DHA-loaded glial cells stimulated with zymosan also release docosanoids, with these novel resolvins possessing such potent anti-inflammatory effects as inhibiting leukocyte trafficking in vivo and proinflammatory cytokine release by stimulated human glial cells. Collectively, the LXs and resolvins represent novel classes of anti-inflammatory agents that are tightly associated with the resolution of acute inflammation and shown to be implicated in the pathogenesis of disease processes, including atherosclerosis, periodontitis, chronic liver disease, and asthma. Moreover, LXs and their analogues are proving to be highly effective therapeutics in a range of experimental disease models, including immune-mediated glomerulonephritis and renal ischemia-reperfusion injury, a range of skin inflammation-like diseases and gastritis. Lipid mediators of this sort are not only natural and essential components of acute inflammatory resolution, but show that when applied to inflammatory disease processes

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are highly effective, thereby providing the rationale for the development of compliant and stable mimetics that target key aspects of chronic inflammation, either ongoing or recurrent, forcing them down a revolving pathway and into remission [11].

## GETTING RID OF LEUKOCYTES

One of the hallmarks of acute inflammation is white blood cell accumulation (PMNs and eosinophils, for instance), designed to neutralize and eliminate the injurious agents. Once the PMNs and eosinophils have done their job and their help is no longer needed, what happens next? At this juncture it must be borne in mind that these are a formidable cell type and if left unchecked could do untold damage to an already inflamed site. After all, these cells are designed to combat infection by releasing hydrolytic and proteolytic enzymes as well as generating reactive oxygen species. Therefore, PMNs and eosinophils must be disposed of in a controlled and effective manner. To oversee this, nature has come up with an ingenious way of defusing such potentially explosive cells called programmed cell death or apoptosis. Apoptosis of inflammatory cells is a physiological process for the nonphlogistic removal of cells. During apoptosis, cells maintain an intact membrane and, therefore, do not release their potentially histotoxic agents. Necrosis of inflammatory leukocytes, on the other hand, involves a loss of membrane integrity leading to the release of potentially toxic intracellular contents [12,13]. Moreover, apoptotic cells express a repertoire of surface molecules that allow their recognition and phagocytosis by macrophages. In fact, the way these cells die helps the resolution process enormously. Recognition of these apoptotic cells by macrophages does not liberate proinflammatory agents from the macrophages themselves but can release anti-inflammatory signals such as IL-10 and TGFβ (endogenous immunosuppressive agents). Thus, not only is apoptosis a noninflammatory way of disposing of cells, but this method has the added advantage of conferring upon macrophages an anti-inflammatory phenotype conducive to resolution and curtailment of ensuing adaptive immune responses. It is important to note that if not recognized and disposed of, apoptotic cells will eventually undergo secondary necrosis releasing damaging intracellular contents and amplifying the inflammatory response. Therefore, increasing the rate of apoptosis, as a potentially anti-inflammatory strategy, must be matched by a mechanism that upregulates macrophage phagocytic clearance capacity. Thus, the removal process might also be susceptible to selective modulation by pharmacological agents for therapeutic gain.

Enhanced undesirable apoptosis occurs in many neurological diseases, such as Alzheimer's disease, Parkinson's disease, Huntington's disease, and multiple sclerosis. Furthermore, inappropriate inflammatory responses or dysfunctional vascular effects leading to tissue damage with increased apoptosis have been observed. So, there is good evidence of cell or tissue apoptosis during myocardial infarction, stroke, or sepsis. Consequently, a therapeutic strategy to delay or inhibit apoptosis would seem a viable option assuming that cell specificity can be achieved. On the other hand, there is much evidence indicating that reduced apoptosis occurs in most cancers. Essentially, uncontrolled cell division or proliferation, there is an apparent failure of cancerous cells to undergo apoptosis. It has also been proposed that in many inflammatory diseases (e.g., rheumatoid arthritis, atopic dermatitis, Crohn's disease, asthma, and chronic obstructive pulmonary disease) there might be delayed apoptosis of key inflammatory cells, thereby prolonging the functional responsiveness of these potential histotoxic cells. A strategy to specifically promote death of cancer cells or tissue-damaging inflammatory cells is therefore likely to be therapeutically beneficial. However, as stated, any attempts to induce cell, especially inflammatory cell, apoptosis must be matched by effective noninflammatory clearance by phagocytic cells (e.g., macrophages). Failure to remove these apoptotic cells may lead to the cells becoming necrotic, thereby increasing the potential for tissue damage. Great progress has been made in recent years in the elucidation of the complex mechanisms that are involved in recognition of apoptotic cells (or apoptotic bodies) by phagocytes (at least 10 recognition mechanisms have been identified so far). Furthermore, phagocytosis of apoptotic cells has been shown to be a highly regulatable process and therefore likely to be amenable to pharmacological manipulation. It has been shown, for example, that elevation of cAMP by PGs can downregulate macrophage capacity to ingest apoptotic cells, whereas treatment of phagocytes with glucocorticoids, LXs (arachidonic acid metabolites via the COX/LOX or LOX/LOX interaction pathways), or even certain cytokines can markedly increase macrophage clearance of apoptotic cells. Interestingly, the environment in which phagocytes are likely to reside can also upregulate apoptotic cell clearance. For example, interactions with extracellular matrix components, such as fibronectin, and ligation of macrophage CD44 with cross-linking antibodies can augment the capacity of phagocytes to engulf apoptotic cells. So, clearance of apoptotic cells in a noninflammatory manner by phagocytes is a therapeutic possibility. Indeed, there is already evidence in animal models that CD44 has an important

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role in resolving lung inflammation and that glucocorticoids might exert some of their therapeutic beneficial anti-inflammatory effects in patients with asthma by influencing apoptosis and apoptotic cell clearance. Some of these novel developments have led to the design of drugs that have even gone into clinical trials. On this note, it is interesting to speculate upon how many clinically used anti-inflammatory drugs trigger hither unknown proresolution pathways in addition to their classic role of dampening conventional proinflammatory events.

Currently, there are no drugs in the clinic that are purposefully based on the elicitation of proresolving pathways with the exception of those drugs that target apoptosis. One extremely active area of drug development that targets apoptosis is the identification of small-molecule caspase inhibitors. Caspases are a family of cysteinyl aspartate–specific proteases that are of fundamental importance in the initiation and execution of apoptosis, ultimately being responsible for the dismantling of the cell during apoptosis. A number of small-molecule inhibitors of caspases (e.g., ZVAD-fmk) have already been tested in animal models of human disease with remarkable success. For example, specific caspase inhibitors have been shown to be effective in preventing or reducing the effects of ischemia (e.g., organ failure and death) in various animal models. These inhibitors of apoptosis are now being tested preclinically or have reached clinical trials for hepatic disease, acute myocardial infarction, and sepsis. A good example of a novel and specific broad-spectrum caspase inhibitor that has been successfully used preclinically is PF-03491390 (formerly named IDN-6556). This compound, administered by a number of routes, was shown to be potent and efficient in reducing signs of liver damage in in vivo rodent models of liver disease. It was shown that the compound seemed to exert its activity by effectively inhibiting caspase activity. In a recent study, the induction of PMN apoptosis during acute inflammation using R-roscovitine (Seliciclib or CYC202), a cyclin-dependent kinase inhibitor (cyclin-dependent kinases override antiapoptotic survival signals from survival factors such as GM-CSF), resulted in enhanced PMN apoptosis and early inflammatory resolution. There are a number of other drugs that are in development for inhibiting apoptosis, but so far the most promising therapeutic development is in the induction of apoptosis, especially for the treatment of a number of cancers. The strategy for inducing apoptosis has been to block powerful survival pathways, for example, by inhibiting BCL2-mediated survival using antisense oligonucleotides, and interference of survival pathways that are mediated by NF-κB, phosphoinositol-3-kinase, and tyrosine kinase activation. Another strategy is to directly induce apoptosis by engaging death receptor pathways (such as FASR, TNFR, and TRAILR) or by other less well-defined mechanisms.

## MACROPHAGES AND RESOLUTION – CELLULAR PLAYERS AND HOMEOSTASIS

From our text book reading the role of the lymphatic system in localized acute inflammation, it is clear that it plays a role in draining inflammatory mediators and effete leukocyte away from the inflamed site. We have already discussed the importance of PMN clearance to the resolution of acute inflammation, but it is equally important that phagocytosing inflammatory macrophages are cleared away from the inflamed site to prevent local macrophage-induced tissue damage and potential granuloma tissue damage and the development of chronic inflammation. However, despite the need to understand the endogenous control of macrophage clearance during acute inflammatory resolution, little is known about this field. There is increasing evidence that macrophage clearance from an inflamed site is a highly regulated event. Using an experimental model of acute resolving peritonitis, it was shown that macrophages adhere specifically to mesothelium overlying draining lymphatics and that their emigration rate is regulated by the state of macrophage activation providing the first evidence that macrophage emigration from the inflamed site is controlled by adhesion molecule regulation of macrophage–mesothelial interactions. This report highlights the importance of adhesion molecules controlling clearance of inflammatory macrophages into the draining lymphatic circulation, thus underscoring new pathways in the resolution of acute inflammation.

Despite the need to clear macrophage from sites of injury, there is emerging evidence that such cells play an important role in eliciting the final phase of inflammation – triggering homeostasis and immune recovery. Macrophages are generally classified as either classically (M1) or alternatively (M2) activated [14]. While this nomenclature is based on the phenotype macrophages acquire in response to defined stimuli in vitro, inflammatory characteristics of macrophages at sites of inflammation in vivo are less well studied. In particular, the phenotype of macrophages found during resolving inflammation is little unknown. Despite this, we have made some advances in understanding this by characterizing the inflammatory nature of macrophages found at the site of resolving peritonitis (Table 2.1). Interestingly, these so-called resolution-phase macrophages, in the context of resolving peritonitis at least, are neither classically nor alternatively activated but are a hybrid of both canonical definitions while they express mannose receptor, synthesize IL-10, and arginase 1 but also express

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**TABLE 2.1. Phenotype of resolution-phase macrophages**

| | Resolving (rM) | Nonresolving (M1) |
|---|---|---|
| Proinflammatory cytokines/chemokines | 0 | +++ |
| Anti-inflammatory cytokines/chemokines | +++ | 0 |
| Bactericidal | + | +++ |
| Mannose receptor | +++ | + |
| HMGB-1 | 0 | +++ |
| iNOS | +++ | + |
| COX-2 | +++ | + |
| cAMP | +++ | + |

M1 macrophage markers, that is, COX-2 and iNOS (Figure 2.2). And while elucidating their precise role in resolution is in its infancy, we are finding that resolution-phase macrophages play a critical role in signaling the influx of innate-type lymphocytes to sites of resolving inflammation. It transpires that as inflammation resolves lymphocytes repopulate the cavity comprising B1, NK, gamma/delta T, CD4+/CD25+, and B2 cells. In particular, repopulating lymphocytes do not bring about resolution but are critical for modulating inflammatory responses to secondary infection and associated mortality (Figure 2.3). While the repopulation of innate-type lymphocytes has been found in several experimental models including mouse and man, the signals that control their postinflammation repopulation and the

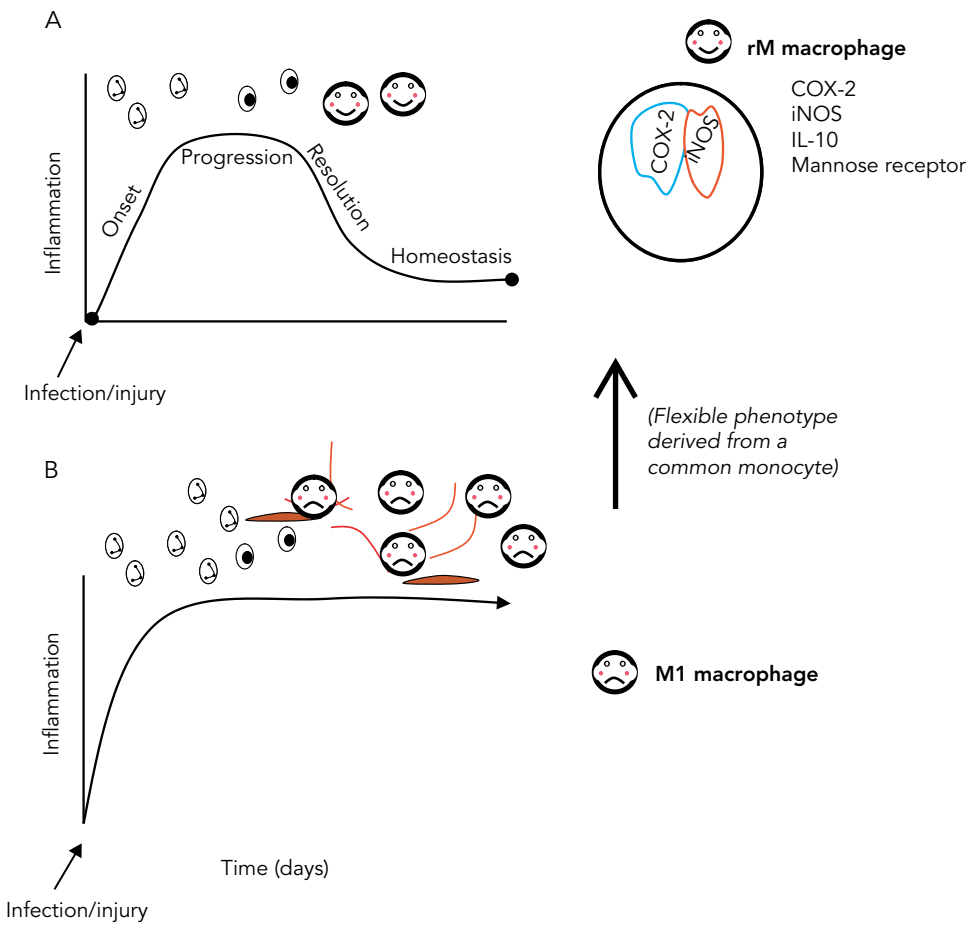



**Figure 2.2.** Inflammatory cell profile and resolution-phase leukocyte phenotype. In response to injury or infection leukocyte migrate to sites of injury. (A) Provided the injurious agent is neutralized inflammation will resolve leading to the injured tissue regaining its prior physiological function/state which we have recently found is characterized by a population of novel macrophages that possess a unique and distinct phenotype, termed rM cells for resolution-phase macrophages. However, (B) if the injurious agent is not cleared or there is failure of proresolution pathways, inflammation will persist characterized by M1 or classically activated proinflammatory macrophages that propagate the response and cause tissue injury. The objective, therefore, is for inflammation to resolve with stromal cells and leukocytes attaining an immunosuppressive "resolution" phenotype.

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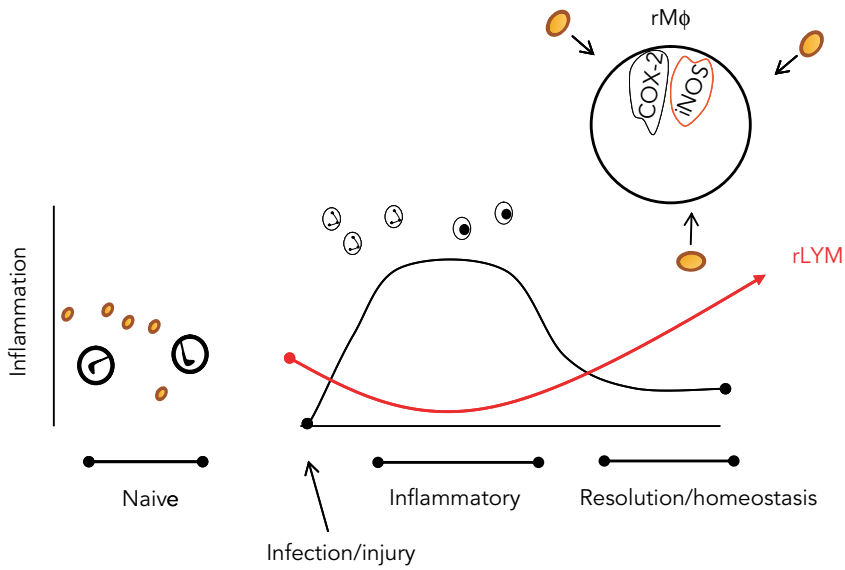



**Figure 2.3.** Resolution leading to homeostasis. One of the objectives of resolution is for inflamed tissues to resolve with limited tissue injury and to reacquire the physiological state it enjoyed before injury. To emphasize this, we need to appreciate that naive tissue have a complement of innate immune cells that control the severity of inflammatory responses. The peritoneal or pleural cavities, for instance, have a defined population of innate-type lymphocytes which control the severity of acute responses as well as tissue-resident macrophages, in a state of immune suppression. Once inflammation ensues, these protective cells transiently disappear and remain absent for the duration of the response only to repopulate tissue postinflammation/resolution. The objective therefore is to achieve the state the tissue enjoyed before injury to restore homeostasis and immune function.

subtypes that confer protection and signal homeostasis are unknown at this stage. Thus, macrophages are important for resolution and restoration of homeostasis after inflammation with the phenotype (M1, M2, or rM) dictating whether inflammation abates or progresses to wound healing. This will depend on the degree of inflammation and associated tissue injury. We certainly know, for instance, that the phenotype of macrophages involved in the progression and resolution of liver injury or chronic renal damage have distinct inflammatory characteristics and that the true phenotype of such protective cells will most likely depend on the tissues involved and may even be species specific. The important point is that while we have gained a great deal of insight into macrophage function based on in vitro cell studies and in response to defined inflammatory stimuli, there is now a greater need to define the inflammatory characteristic of macrophages taken directly from resolving tissues. Taking this further, I would suggest that in addition to defining resolution as the active clearance of leukocytes, we need to include macrophage phenotype and innate-type lymphocyte repopulation. This will become important when developing drugs with the objective of bringing about resolution. The indices for resolution, therefore, should not only include proinflammatory mediator catabolism and leukocyte clearance but also consider including macrophage-resolution phenotype and innate-type lymphocyte repopulation.

## RESOLUTION OF ADAPTIVE IMMUNE RESPONSES

In the previous sections we have discussed how resolution of innate inflammation resolves, discussing the fate of inflammatory PMNs, eosinophils, and macrophages. However, adaptive immunity, specifically, a Type III hypersensitivity (Arthus reaction) (discussed in the Glossary) or a Type IV delayed type hypersensitivity (DTH), are also acute in nature and resolve in a matter of hours or days. For instance, in a purified protein derivative–induced DTH response, it was shown that the induction and resolution of this response may depend on the expression of cytokines, such as IL-2 and IL-15, which regulate both proliferation and apoptosis in T cells. Failure to control either of these phases of the reaction may contribute to the chronicity of T lymphocyte–mediated inflammatory reactions. In another important series of studies, the endogenous factors that control the longevity of

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granulomatous autoimmune thyroiditis revealed that the ratio of CD4+/CD8+ T cells are critical determinants of its resolution. In this disease process, CD4+ T cells outnumber CD8+ T cells when lesions progress to fibrosis, while CD8+ T cells outnumbered CD4+ T cells in thyroids that resolved.

## WOUND HEALING

Importantly, successful resolution will limit excessive tissue injury and give little opportunity for the development of chronic, immune-mediated inflammation. However, if the host is unable to neutralize the injurious agent and/or there is a failure of endogenous proresolving mediators to invoke resolution, then acute inflammation might perpetuate, resulting in varying degrees of tissue injury. If tissue injury is mild, necrotic parenchymal cells will be replaced by new cells of the same type in a process known as regeneration. If, however, tissue damage is extensive, or when fibrin is not rapidly cleared after acute inflammation, the process of healing is by repair. This involves the in-growth from the surrounding connective tissue of an initially vascular tissue containing capillary loops, fibroblasts, and leukocytes, and is known as granulation tissue. With time, the fibroblasts lay down collagen and the capillaries disappear, leaving an avascular area of fibrosis or scar. Repair by granulation and fibrosis occurs in many parts of the body where a deposit of clot, exudate, or dead tissue occurs and is given the general name of organization. Undoubtedly, this is also a form of inflammatory resolution associated with tissue damage. As in rheumatoid arthritis and asthma, for example, there will be continuous or repeated bouts of acute inflammation, resulting in ongoing tissue damage. Attempts at wound healing would result in granulomatous tissue formation, angiogenesis, fibrosis, and scar formation, all occurring concurrently. This is chronic inflammation and might be defined as a continuous inflammatory disease state that could be driven by the development of an immune response to an endogenous antigen (autoimmunity).

## CONCLUSION

The take-home point from all these studies is that regardless of the etiology of the inflammatory response, if it is resolving in nature then this resolving event is highly controlled and regulated by endogenous factors and mechanisms that, if interfered with, may lead to chronic inflammation and autoimmunity. Therefore, this emerging concept of resolution presents with drug discovery opportunities. For instance, we can develop drugs that mimic endogenous proresolution factors or enhance their synthesis. By analogy, just like IL-1β and TNFα, which are expressed during the early onset phase of acute inflammation, can be used to initiate acute inflammation experimentally, then factors expressed during resolution may be useful in switching off an ongoing inflammatory response. Alternatively, we can develop drugs that enhance the synthesis of endogenous proresolution factors. Equally, we can develop drugs or refine existing ones (e.g., glucocorticoids) to bring about selective leukocyte apoptosis (eosinophil) while at the same time enhance their phagocytosis. If these aims seem too unrealistic at present, then for the foreseeable future perhaps we should bear in mind that when developing novel anti-inflammatories such drugs should not interfere with the synthesis or mode of action of proresolution mediators and/or the mechanisms that are critical for resolution. Finally, from a basic concept perspective this chapter aimed at underscoring the complexity of acute inflammation and the fact that while there are endogenous factor that drive the response, there are also factors produced by inflammatory cells and injured tissues that counterbalance the degree of inflammation and prevent the response from causing unnecessary tissue injury. After all, inflammation is a good thing but it has the potential to do great harm. Therefore, not only the magnitude of the initial response to injury/infection must be tightly controlled, but also its longevity and the factors that switch it off.

### GLOSSARY OF TERMS

**Polymorphonuclear leukocyte** Polymorphonuclear leukocytes (PMNs) were discovered by Paul Ehrlich who used contemporary fixing and staining techniques to identify the lobulated nucleus and the granules that typify cells that we now classify as eosinophils, basophils, and PMNs.

**Vascular leakage** The process of the escape of plasma and plasma proteins along with white blood cells from the vessel is known as exudation. This inflammatory exudate accounts for an increase in the volume of interstitial fluid (edema) and tissue swelling at the local site of injury.

**Peroxisome proliferator-activated receptor** Peroxisome proliferator-activated receptors (PPARs) are members of the nuclear receptor family that regulate the transcription of genes involved in lipid and lipoprotein metabolism, glucose and energy homeostasis, as well as cellular differentiation and consist of three isotypes, alpha (NR1C1), gamma (NR1C3), and beta/delta (NRC1C2) with a differential tissue distribution.

**NF-κB** Nuclear factor kappa B (NF-κB) is a group of sequence-specific transcription factors that are best known as a key regulator of the innate and adaptive inflammatory responses, cell survival, and oncogenesis. In mammals, NF-κB consists of five structurally related and functionally conserved proteins, RelA (p65), RelB, c-Rel, NF-κB1 (p105 and p50), and NF-κB2 (p100 and p52).

**Nonphlogistic** Noninflammatory. This term is used to describe the clearance of leukocytes in a manner that does not elicit an inflammatory response.

**Hypersensitivity reactions** A delayed type hypersensitivity or Type IV hypersensitivity is mediated by T lymphocytes and not by antibody–antigen complexes (Arthus or Type III hypersensitivity). Typically, this response occurs 24–72

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hours after the sensitized host is exposed to the offending antigen. For example, a DTH reaction may be set up experimentally by sensitizing to methylated bovine serum albumin in Freund's complete adjuvant and challenging 12 days later with the same antigen.

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# 3 Links between Innate and Adaptive Immunity

*Christopher L. Karp*

## THE INNATE AND ADAPTIVE IMMUNE SYSTEMS: DEFINITIONS, CONTEXT, AND CONTRASTS

Standard accounts of the immune system emphasize the antigen-specific immunity and memory afforded by the adaptive immune system, contrasting it with the "nonspecific" defenses provided by the phylogenetically more ancient innate immune system. While study of the innate immune system has undergone a recent renaissance, most immunology textbooks still present innate immunity as an initial stopgap defense that holds the line until the "real" (efficient, effective, sophisticated) adaptive immune system can take over. There are obvious flaws in such formulations. First, while adaptive immunity may usefully be seen as a single system – with its cells (B and T lymphocytes) and antigen receptors (immunoglobulins [Ig], T-cell receptors) depending directly on the evolution of the recombination-activating gene (*RAG*) in jawed vertebrates – innate immunity, present in all metazoans, is a congeries of pathways. "Innate immune systems" is a much better term. Second, the innate immune systems are in no way less sophisticated than the adaptive immune system, having been under evolutionary pressure for far longer. Third, the innate immune systems are not of secondary importance; the adaptive immune system is directly dependent on the former for efficient and appropriate activation. Fourth, innate immune effector mechanisms are not less effective than adaptive immune effector mechanisms. (As an example, chemotherapy-induced ablation of neutrophils leads to a high risk of fatal infection with otherwise harmless commensal flora in a telescoped time frame compared with the similar risk of infectious mortality attendant on lymphocyte dysfunction, e.g., in infants with severe combined immunodeficiency.) Fifth, the standard superficial view of the kinetics of innate and adaptive immune responses, the former handing off to the latter, is incorrect; the innate immune systems do not become quiescent when the adaptive immune system is activated.

The divide between those studying innate (previously "cellular") and adaptive (previously "humoral") immunity dates back to the earliest days of immunology as a science. The ascendancy of those studying adaptive immunity in the field over the past several decades is something of a sociological/historical accident, albeit fueled in part by the compelling nature of the problems being investigated (e.g., the generation of lymphocyte receptor diversity, the nature of tolerance to self, the biology of functional polarization of effector and regulatory cell types) and by the mechanistic insights thereby obtained. The recent molecular identification of key innate immune receptors that signal the presence of microbial products has revitalized the study of innate immunity; the pendulum is swinging back. But, in many ways, the distinction between innate and adaptive immunity is an artificial one. The two are inextricably linked in vertebrates – something that forms the focus of this chapter.

Prior to outlining these links (an outline that will, perforce, be illustrative as opposed to comprehensive), it is useful to sketch out and contrast innate and adaptive immunity. The essential function of these immune systems includes protection against the microbial universe and injury (broadly defined). These functions can usefully be broken down into (a) discrimination of microbes or injury (immune recognition/activation), (b) containment or elimination of microbes or injury (immune effector responses), and (c) control of immune response vigor and duration (counter-regulation, resolution) – all of which needs to be done without harming the host itself. It should also be noted that, apart from this, the immune systems also play critical roles in development and homeostasis.

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## Immune Recognition

Adaptive immune recognition is a function of receptors that are clonally distributed on individual T cells (T-cell receptors) and B cells (B-cell receptors: surface Ig), or secreted by the latter (Ig). The receptor specificities of T and B cells are essentially infinite, this repertoire being generated through somatic rearrangement of gene segments and somatic mutation. Specific receptor–bearing lymphocytes are selected for (or against) during the development of each individual organism. The ability of the adaptive immune system to generate receptors that can recognize any molecular pattern has consequences for self-recognition. Deleterious recognition of self by the adaptive immune system is largely avoided by diverse mechanisms collectively referred to as "immunological tolerance," including developmental selection against lymphocytes bearing receptors with inappropriate affinity for self-antigens; alteration of the function of lymphocytes that encounter (self-) antigens in the absence of cues from the innate immune system (*vide infra*); and active suppression of (perniciously) autoreactive lymphocytes by specialized populations of regulatory lymphocytes.

Immune recognition in the innate immune system is quite different. For one, the receptors are germ line–encoded and non-rearranging (with the exception of isoform generation). This means, perforce, that innate immune receptors are relatively few in number, and that receptor repertoire selection reflects evolutionary processes. In 1989, Charles Janeway wrote a landmark theoretical paper outlining an elegant framework for understanding these constraints on innate immune recognition, conceived as the discrimination of noninfectious self from infectious nonself. Given the enormous molecular variability and high mutation rate of microorganisms, he postulated that (a) the molecular structures recognized must be shared by large groups of pathogens, (b) such structures must be tightly constrained from mutational variation by being essential to microbial survival, and (c) such structures must be completely distinct from host structures. In this schema, recognition of microbial nonself occurs through pattern recognition receptors (PRRs) that bind to pathogen-associated molecular patterns (PAMPs; something of a misnomer: the structures recognized by PRRs are not unique to pathogens). The subsequent discovery in 1998 of the Toll-like receptor (TLR) family of membrane-bound PRRs that signal in response to conserved microbial products (Chapter 13) not only appeared to fit this schema beautifully, but, more generally, revitalized study of the innate immune system. This led, quite rapidly, to discovery and/or delineation of other PRR families, including the NOD-like receptors (NLRs) and RIG-I-like helicase receptors (RLRs). A competing theoretical framework for understanding innate immune recognition, the danger hypothesis, was put forward by Polly Matzinger in the early 1990s. In this model, the primary driving force for the immune system and immune recognition is not self/nonself discrimination but protection against, and hence detection of, danger. Obviously, the presence of microbial products in normally sterile sites represents danger. But nonmicrobial (e.g., trauma, tissue ischemia and infarction, crystal deposition) danger abounds, and also leads to innate immune activation, suggesting that there must be receptors for structures induced, upregulated, or solubilized by cellular injury. The fact that sterile inflammation often mirrors microbe-driven inflammation suggests likely overlap between PAMPs and receptors for altered or injured self damage-associated molecular patterns (DAMPs), something that has been borne out experimentally. Finally, it should be noted that, despite the desire for theoretical simplicity, evolution works in an ad hoc fashion. It is thus not surprising that there is at least one other mode of innate immune recognition that does not fit easily into either the microbial nonself or the danger rubric (theoretical constructs that are, in any case, not mutually exclusive) – recognition of missing self. In this mode, seen with both natural killer (NK) cells (Chapter 8) and the alternative pathway of complement activation, the engagement of molecular structures only expressed by normal host cells (major histocompatibility complex [MHC] class I and regulators of complement activation family members, respectively) inhibits activation. An overview of innate immune receptors is provided in Table 3.1.

## Immune Effectors

B cells and α/β T cells (including CD8+ and CD4+ T cells) are the effector cells of the adaptive immune system. In terms of the effector pathways mediated by such cells, B cells produce Ig, the binding of which to antigens, microbes, and/or host-derived structures can facilitate phagocytosis ("opsonization"), activate complement, activate or inhibit the activation of cells bearing receptors for the Fc portion of Ig, and/or alter the function of the structures so bound (e.g., neutralization of viruses and microbial toxins). Like all immune cells, B cells also produce a variety of secreted protein mediators (cytokines) that act in autocrine, paracrine, and systemic fashion to alter the function of other cells, regulating the activity of other immune effector cells and controlling inflammatory responses. The effector mechanisms of CD4+ and CD8+ T cells include diverse pathways of inducing apoptosis ("cytolysis") as well as the regulation of inflammation and immune responses via cytokine production and cell surface molecules.

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**TABLE 3.1. Pattern recognition receptors of the innate immune systems**

| Receptor class/receptor | Location | Ligands | Comments |
|---|---|---|---|
| **Toll-like receptors (TLRs)**<br>(TLR1–13) | TLR1,2,4,5,6,10–13: plasma membrane; TLR3,7,8,9: endosomal compartments | Diverse PAMPs and DAMPs (Chapter 13) | Activation of NF-κB, MAP kinase, and IRF pathways |
| **NOD-like receptors (NLRs)**<br>NALP1–14<br>NOD1–5<br>NAIP, IPAF, CIITA | Cytoplasmic | Diverse PAMPs and DAMPs | Activation of NF-κB; activation of caspase 1 leading to IL-1β cleavage or cell death |
| **RIG-I-like helicase receptors (RLRs)**<br>RIG-I<br>MDA5 | Cytoplasmic | Viral RNA: RIG-I: 5′ triphosphate ssRNA; MDA5: dsRNA | Activation of NF-κB and IRF pathways |
| **DAI** | Cytoplasmic | dsDNA from pathogens, damaged host cells | Activation of NF-κB and IRF pathways |
| **PKR** | Cytoplasmic | dsRNA | Serine/threonine kinase: phosphorylation of eIF2α, blocking protein synthesis; activation of NF-κB and MAP kinase pathways |
| **2′-5′-Oligoadenylate synthase** | Cytoplasmic | dsRNA | Activation of RNaseL: RNA degradation |
| **Dectin-1** | Plasma membrane | β-Glucans from fungi and other microbes | C-type lectin; cooperation with TLR2; activation of NF-κB and MAP kinase pathways; phagocytosis |
| **Phagocytic receptors**<br>numerous, including<br>Scavenger receptors (class 1–6)<br>MΦ mannose receptor<br>DEC-205<br>DC-SIGN<br>Langerin<br>CR3, CR4 | Plasma membranes of macrophages and/or dendritic cells (DCs) | Diverse microbial and altered host components | Multiple protein families; expressed on the surface of professional phagocytes: macrophages, DCs, neutrophils; those on DCs are important for the delivery of antigen to processing compartments |
| **Secreted PRRs**<br>numerous, including | | | |
| Lipid transferases | | | |
| BPI | | Lipid A | Killing/opsonization of Gram-bacteria transfer of LPS to CD14 |
| LBP | | Lipid A | |
| Collectins | | | |
| MBL | | Microbial sugars | Complement activation, opsonization |
| SP-A | | Diverse pathogens | Opsonization |
| SP-D | | Diverse pathogens | Opsonization |
| Pentraxins | | | |
| C-reactive protein | | | Opsonization, complement activation |
| Serum amyloid P | | | Opsonization, complement activation |
| PTX3 | | | Opsonization, complement activation |
| PGRPs | | Peptidoglycan | Bacterial killing |

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| Receptor class/receptor | Location | Ligands | Comments |
|---|---|---|---|
| fMLR | Plasma membrane, neutrophils | N′-formylated peptides | G-protein–coupled receptor; chemotaxis, activation |
| Activating NK cell receptors<br>numerous, including<br>Activating KIRs<br>NKG2D<br>NKp46, 44, 30 | Plasma membrane | Diverse viral, and stress-inducible host ligands | |

*Note*: Both this and the following table, like the chapter as a whole, are illustrative as opposed to comprehensive. BPI, bactericidal permeability-increasing protein; CR, complement receptor; LBP, LPS-binding protein; MBL, mannose-binding lectin; PGRP, peptidoglycan receptor protein; SP, surfactant protein.

The innate immune system consists of an array of specialized immune cells: monocytic cells (including monocytes, macrophages, and most dendritic cells [DCs]), other DC subtypes (such as plasmacytoid DCs, whose origin remain somewhat unclear), granulocytes (including neutrophils, eosinophils, mast cells, and basophils), and innate lymphocytes (including NK cells, NKT cells, γ/δ T cells, and "innate" B cells). Innate immunity comprises more than just specialized immune cells, however. Epithelial cells lining body surfaces are central to innate immunity: acting as barriers, producing antimicrobial effectors, secreting substances that prevent microbial attachment and entry, and secreting mediators that regulate inflammation. Endothelial cells are similarly central to regulation of both innate and adaptive immunity, regulating immune cell trafficking into tissues (Chapter 11). Hepatocytes play essential roles through the secretion of humoral receptors and effectors. More broadly, the presence of intracellular sensors and effectors that respond to viral nucleic acids in all nucleated cells suggests that, at a fundamental level, all cells are part of the innate immune systems. An overview of innate immune effector pathways is provided in Table 3.2. It will be noted that, in addition to cell-associated effector pathways, a variety of humoral effector pathways (including proteolytic cascades such as the complement and coagulation systems, and secreted PRRs) are part of the innate immune systems. Finally, the induction and regulation of local and systemic inflammation is, in large part, a function of the production of mediators by innate immune cells.

## Immune Counter-Regulation

While inflammatory responses are critical for protection against the microbial universe (and danger), all inflammatory responses have the potential for harming the host. Indeed, it has become clear that dysregulated inflammation is central to the pathogenesis of a wide spectrum of diseases, including infectious (e.g., sepsis), autoimmune (e.g., multiple sclerosis), autoinflammatory (e.g., juvenile rheumatoid arthritis), allergic (e.g., allergic asthma), vascular (e.g., atherosclerosis), neurodegenerative (e.g., Alzheimer's disease), metabolic (e.g., type II diabetes and other metabolic sequelae of obesity), and monogenic (e.g., cystic fibrosis) diseases. As a result, immune responses must be tightly regulated in space, time, amplitude, and character. While immunologists have traditionally focused on the molecular mechanisms underlying immune activation and class specification, control of the amplitude and resolution of immune responses is just as important (Chapter 2). In recent years, numerous, often overlapping and redundant immune counter-regulatory mechanisms that control innate and/or adaptive immune responses have been defined, including specialized cells (e.g., Foxp3+ regulatory T cells), cytokines (e.g., IL-10), enzymes (e.g., indolamine 2,3-dioxygenase [IDO]), cell surface receptors (e.g., CTLA-4), intracellular signaling modulators (e.g., SOCS proteins), lipid mediators (e.g., lipoxins), and cellular reprogramming (e.g., endotoxin tolerance).

## Relationship between Acute versus Chronic Inflammation and Innate versus Adaptive Immunity

What is the relationship between acute and chronic inflammation, defined and described in Chapter 1, and the innate and adaptive immune systems? Classic, acute (neutrophilic) inflammation – as seen early in pyogenic infection and with tissue infarction – represents an innate immune response (albeit likely under the control of counter-regulatory mechanisms deriving from both the innate and adaptive immune systems). Chronic (lymphocytic, monocytic) inflammation – as seen in delayed type hypersensitivity and granulomatous responses – represents coordinate activation of both the innate and adaptive immune systems. It should be noted, however, that classical "acute" and "chronic" inflammation represent only small portions of the spectrum of inflammatory responses mounted by the human (and murine) innate and adaptive immune systems. For example, many of the disease-associated

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**TABLE 3.2. Effector mechanisms of the innate immune systems**

| System or mechanism | Functions |
|---|---|
| Epithelia | Barrier function<br>Secretion of antimicrobial peptides and proteins<br>Secretion of iron-binding proteins<br>Secretion of mucins (mucosae; inhibition of microbial attachment)<br>Mucociliary clearance (airway epithelia)<br>Secretion of opsonins (including those activating complement)<br>Secretion of chemotactic cytokines and lipid mediators<br>Secretion of proinflammatory cytokines<br>Secretion of complement components |
| Phagocytosis (macrophages, DCs, neutrophils) | Intracellular killing via<br>Antimicrobial peptides<br>Hydrolytic enzymes<br>Reactive oxygen species<br>Reactive nitrogen species<br>Nutrient competition<br>Antigen processing and presentation |
| Complement system | Opsonization<br>Direct antimicrobial activity<br>Regulation of inflammation, immunoregulation |
| Cytolysis | Induction of apoptosis in virally infected and stressed cells by NK cells (via perforin/granzyme, via surface-expressed TNF family members) and by IFN-α/β-inducible gene products |
| Noncytolytic antiviral effectors | Induction of apoptosis (above), inhibition of viral transcription, replication, assembly (Mx proteins, GBP proteins); blockade of protein synthesis (PKR); RNA degradation (OAS)<br>Inhibition of viral gene expression and replication by IFN-γ, TNFα |
| Extracellular killing | Neutrophils (via extracellular "nets")<br>Basophils, eosinophils, mast cells (defense against helminths) |
| Hepatocytes | Proinflammatory cytokine-driven production of "acute phase proteins": opsonins, complement components |
| Coagulation system | Walling off of infected tissues, regulation of inflammation |

inflammatory responses noted earlier – such as allergic inflammation (involving innate and adaptive immunity); the chronic, low-grade inflammatory state associated with obesity (apparently, largely innate); the neurotoxic inflammation associated with Alzheimer's disease (also largely innate) – fit poorly, if at all, into these categories.

## REGULATION OF ADAPTIVE IMMUNITY BY THE INNATE IMMUNE SYSTEM

At a fundamental level, efficient activation of the adaptive immune system is dependent on innate immunity – while the adaptive immune system has the ability to respond to essentially any molecular structure, it relies on the innate immune system to discriminate what structures should be responded to.

### Activation of Adaptive Immune Responses: T Cells

It has been known for 40 years that effective lymphocytic responses to antigens do not occur in the absence of mononuclear phagocytic cells. In contemporary terms, the generation of effector T cells from naive CD4+ and CD8+ T cells is dependent on antigen presentation by a class of innate immune cells: DCs. DC presentation of peptides on MHC class I and II molecules to T-cell receptors (TCRs) on naive CD4+ and CD8+ T cells, respectively, with appropriate affinity for the specific peptide/MHC complex ("signal 1") is necessary, but not sufficient, for T-cell activation, clonal expansion, and the generation of effector and memory cells. In the absence of costimulation, the provision of signals ("signal 2") through cell surface–expressed molecules on the APC (e.g., CD80 and CD86, CD40) to cognate receptors on the T cell (CD28, CD40L), the result of such antigen presentation is deletion or suppression. DCs (which come in a variety of subtypes, with somewhat different functions) are prodigious samplers of their antigenic environment, both external and internal. In part, external sampling is a function of phagocytosis induced by innate immune PRRs (e.g., complement receptors and scavenger receptors); in part, this is due to fluid phase sampling via macropinocytosis. Effective antigen presentation depends on

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DC maturation (antigen processing, upregulation and loading of MHC class I and II complexes, upregulation of costimulatory molecule expression) and migration to the T-cell areas of lymph nodes. In turn, such maturation is a result of signaling through innate immune PRRs.

This necessity for DC maturation provides a mechanistic explanation for adjuvanticity – what Janeway called the "immunologists' dirty little secret." Purified, soluble proteins on their own fail to drive adaptive immune responses. The ability of bacterial products to act as adjuvants has long been appreciated. Lipopolysaccharide (LPS), acting through TLR4, is a paradigmatic case in point. TLRs are clearly not the only PRRs that drive such DC activation and maturation; both microbial and endogenous danger signals can suffice. For example, aluminum hydroxide (alum), the adjuvant most widely used clinically, now appears to act through induction of uric acid, a DAMP that signals through the NLR, Nalp3. While the key molecular details of PRR-driven DC maturation remain somewhat unclear, the type I IFNs (IFN-α[s] and -β) appear to be of special importance. A variety of other stimuli are also able to drive DC maturation, including interactions with other innate immune cells (NK cells and NKT cells, whose production of IFN-γ after PRR engagement can drive DC maturation in a paracrine fashion), neutrophils, and mucosal epithelia. Cells rendered apoptotic by NK cells can drive robust adaptive immune responses, a phenomenon dependent on signaling pathways downstream of the TLRs. An overview of innate immune regulation of T-cell activation is presented in Figure 3.1.

While the generation of effector T cells from naive T cells appears to depend largely on antigen presentation by DCs in lymph nodes, the *reactivation* of memory cells in tissues can occur efficiently via other antigen-presenting cells (APCs), among which cells of the macrophage lineage appear to play a dominant role. Further, macrophages (along with other innate immune cell types and epithelial cells) act as critical, tissue-specific sensors of infection and injury, secreting chemotactic cytokines that drive and regulate not only innate immune inflammation, but also the recruitment and activation of DCs and lymphocyte populations.

### Activation of Adaptive Immune Responses: B Cells

Efficient activation of most B-cell responses is also dependent, either directly or indirectly, on the innate

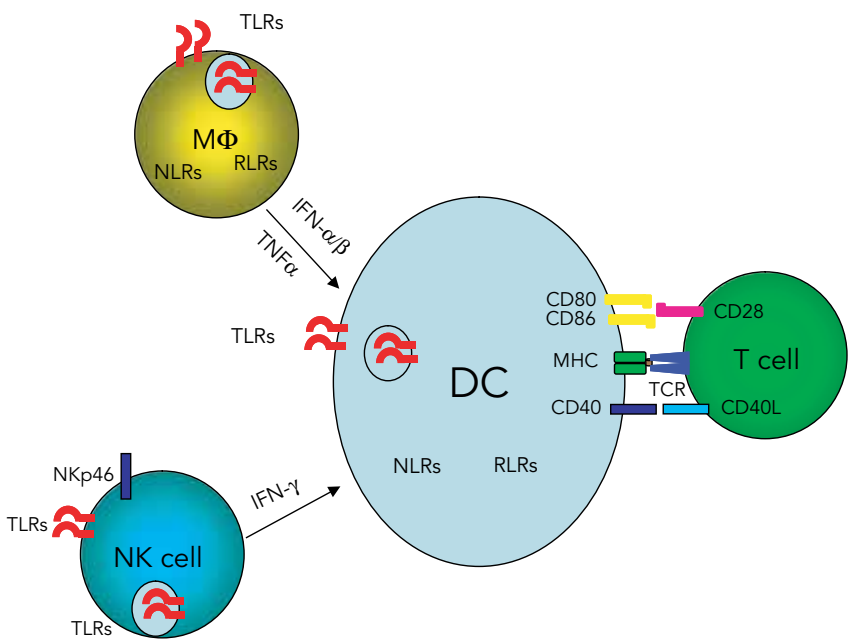



**Figure 3.1.** Innate immune regulation of T-cell activation. DC presentation of peptides on MHC molecules to T-cell receptors (TCRs) on naive T cells ("signal 1") is necessary, but not sufficient, for T-cell activation, clonal expansion and the generation of effector and memory cells. In the absence of costimulation, the provision of signals ("signal 2") through cell surface–expressed molecules on the APC (e.g., CD80 and CD86, CD40) to cognate receptors on the T cell (CD28, CD40L), the result of antigen presentation is deletion or suppression. "Effective" antigen presentation to T cells is the result of DC maturation: upregulating antigen processing, MHC class I and II expression and loading, and costimulatory molecule expression. DC maturation is a result of PAMP and DAMP signaling through innate immune PRRs (e.g., TLRs, NLRs, RLRs) on DCs. Maturation can also occur through interactions with other innate immune cells. IFN-γ secretion by NK cells (pictured; activated through widely expressed PRRs such as TLRs, or through NK activating receptors such as NKp46) or other innate lymphocyte populations can drive DC maturation; as can TNFα and type I IFN secretion by macrophages activated through PRRs. MΦ, macrophage; MHC, major histocompatibility complex; NLR, NOD-like receptor; RLR, RIG-I-like helicase receptor; TCR, T-cell receptor.

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immune systems. Cognate antigen-induced clustering of the B-cell receptor (BCR, surface Ig) triggers B-cell activation. The presence of complement activation fragments (C3d) on the antigen lowers the threshold for B-cell activation by 1000-fold, through cross-linking of complement receptor 2 (CR2) and associated molecules (CD19, CD81) to the BCR and its associated signaling molecules (Igα and Igβ) (Figure 3.2). Beyond the role of complement, B-cell responses to protein antigens require CD4+ T cell help – the provision of membrane-bound (CD40L) and paracrine (cytokines) signals by antigen-specific CD4+ T cells that stimulate B-cell proliferation (clonal expansion) as well as Ig isotype switching, synthesis, and secretion. As noted earlier, the generation of such helper T cells is itself dependent on antigen presentation by DCs, cells that can also directly present antigen and provide stimulation to B cells through CD40/CD40L interactions.

### Immune Class Specification

In addition to activation and clonal expansion, T cells that avoid deletion in the aftermath of antigen presentation by DCs undergo variable degrees of differentiation down pathways of functional polarization, leading to the generation of cells with diverse effector and/or regulatory properties. The molecular details are best understood for CD4+ T cells, which can differentiate down a variety of pathways that promote specific classes of immune responses. Such polarized CD4+ T-cell populations include **Th1 cells** (producing IFN-γ, among other cytokines; facilitating systemic immunity to intracellular pathogens through activation of the microbial effector pathways of macrophages, generation of complement-fixing Ig isotypes, and activation of CD8+ cytolytic T cells and innate lymphocyte populations; implicated as well in organ-specific autoimmune diseases); **Th2 cells** (producing cytokines such as IL-4, IL-13, IL-5, and IL-9; facilitating cellular and humoral responses to helminths; also implicated in allergic disease and pathological fibrotic responses); **Th3 cells** (producing transforming growth factor [TGF-β], among other cytokines; facilitating mucosal immunity, providing immune counter-regulation); **Th17 cells** (producing IL-17, among other cytokines; facilitating the mobilization of neutrophils to epithelial barriers; also implicated in organ-specific autoimmune diseases); and various populations of regulatory cells, such as **induced Foxp3+ regulatory T cells** ("Treg"; producing cytokines such as IL-10 and TGF-β; suppressing both innate and adaptive immune responses via diverse mechanisms). Control of the functional polarization of CD4+ (and CD8+) T cells is under the instructive guidance of cytokines ("signal 3") produced by different DC subsets (activated under different conditions) as well as by other innate immune cells – the production of which is driven by innate immune PRRs. Innate immune regulation of CD4+ T-cell differentiation and polarization is schematized in Figure 3.3.

In turn, CD4+ T-cell polarization leads to polarization of B-cell Ig isotype production. Help from Th1 cells drives the synthesis of complement-fixing Ig isotypes; that from Th2 cells driving IgE synthesis; and that from Th3 cells driving IgA synthesis.

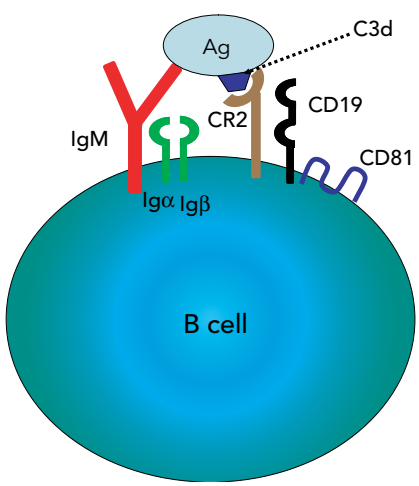



**Figure 3.2.** Innate immune coactivation of B cells. Antigen-induced clustering of the B-cell receptor (BCR, surface Ig) triggers B-cell activation. Decoration of antigen with the complement activation fragment, C3d, lowers the threshold for B-cell activation by a factor of 1,000 via cross-linking of complement receptor 2 (CR2) and its associated molecules, CD19 and CD81, with the BCR and its associated molecules (Igα and Igβ) – leading to the coordinate activation of signaling cascades from both the BCR and the CR2 complexes. In addition to the effects of complement, the innate immune systems play a critical role in B-cell responses to protein antigens. Such responses demand help from antigen-specific CD4+ T cells, the activation of which is dependent on DCs (cells that can also directly present antigen, and provide stimulation to B cells through CD40/CD40L interactions).

### Counter-regulation of Adaptive Immune Responses

In addition to activation and class specification, the innate immune systems play a central role in regulation of the amplitude and resolution of adaptive immune responses. Numerous pathways have been implicated. Among counter-regulatory cytokines, IL-10 appears to play a uniquely important role, restraining the vigor of inflammatory responses: local and systemic, innate and adaptive, and polarized along diverse axes of effector response. IL-10 is produced by a plethora of cell types including diverse innate (myeloid cells, innate lymphocytes, epithelial cells) and adaptive immune cells. It appears likely that the more polarized a B or T cell is, the more likely it is to produce IL-10 for immune counter-regulation. As noted earlier, the generation of such polarization (including the generation of specific classes of counter-regulatory

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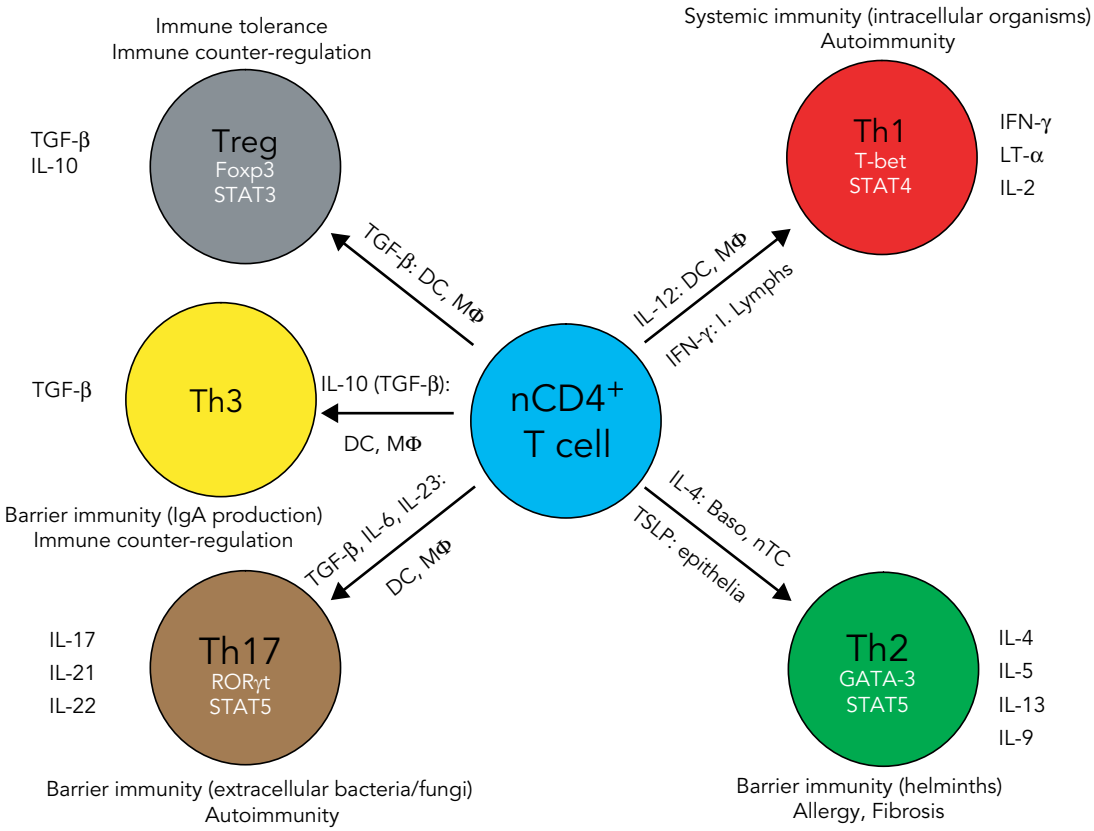



**Figure 3.3.** Regulation of CD4+ T-cell polarization by the innate immune systems. Specification of the effector (or regulatory) class of immune responses by CD4+ T cells is largely the result of functional polarization of naive CD4+ T cells in the periphery into differentiated populations that produce unique patterns of cytokines and have distinctive biological properties. Such polarized CD4+ T-cell populations include Th1 cells, Th2 cells, Th3 cells, Th17 cells, and induced regulatory T cells (Treg; *N.B.,* natural Treg are already differentiated upon leaving the thymus). Control of the functional polarization of CD4+ (and CD8+) T cells is under the instructive guidance of cytokines ("signal 3"), largely derived from DCs, as well as other innate (as well as adaptive) immune cells. The relevant "signal 3," its sources, and the transcription factors (the latter in white, in the cells) critical for the differentiation of these polarized populations, along with their signature cytokine profiles and biological activities, is schematized here. It appears that the more polarized a T cell, the more likely it is to express IL-10 and have counter-regulatory properties. Baso, basophils; I. Lymphs, innate lymphocytes; LT, lymphotoxin; MΦ, macrophages; nTC, naive T cells; TSLP, thymic stromal lymphopoietin.

cells such as Tregs) is largely specified by the innate immune systems. In addition to other inhibitory cytokines (e.g., TGF-β), the innate immune systems provide counter-regulation to adaptive immune responses through inhibitory enzymes (e.g., inducible nitric oxide synthase and IDO), lipid mediators (e.g., lipoxins and resolvins), cell surface molecules (e.g., CTLA-4 and membrane-bound TGF-β); antigen destruction and clearance (e.g., by phagocytic or cytolytic destruction of microbial pathogens, and by complement receptor–mediated clearance of immune complexes); and killing of APCs (e.g., by NK cell–mediated killing of DCs).

## REGULATION OF INNATE IMMUNITY BY THE ADAPTIVE IMMUNE SYSTEM

If the innate immune systems are essential for adaptive immune function, the reverse is no less true. T cells regulate inflammation, in part, through regulation of innate immune cell differentiation state, recruitment, activation, and homeostasis. Innate immune activation is also under counter-regulatory control, both direct and indirect, by the adaptive immune system.

### Regulation of Innate Immune Cell Function

The generation of specific classes of inflammatory responses by the adaptive immune system involves the selective marshaling of innate immune cell functions (along with adaptive immune functions). Again, this has been best worked out for CD4+ T cell–driven immunoregulation, illustrated by the effects of cytokines produced by polarized CD4+ effector T cells on macrophage populations. Th1 cell production of IFN-γ drives antigen-specific, "classical activation" of macrophages, marked by upregulation of microbicidal effector functions (e.g., production of reactive oxygen

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species and nitric oxide) and secretion of proinflammatory cytokines. On the other hand, Th2 cell production of IL-4 and IL-13 leads to "alternative activation" of macrophages, marked by upregulation of repair functions (e.g., driving collagen production and cellular proliferation). More generally, these and other cytokines produced by polarized CD4+ effector T cells have wide effects across the innate immune systems. For example, IFN-γ affects neutrophil function (facilitating recruitment, inhibiting apoptosis, priming for cytokine and chemokine production, and the generation of reactive oxygen species), epithelia (altering chemokine production by airway epithelia) and endothelia (regulating adhesion molecule expression); IL-4 and IL-13 alter the properties of epithelia (driving mucous metaplasia, altering chemokine production) and endothelia (regulating adhesion molecule expression); other Th2-derived cytokines regulate the differentiation, recruitment, apoptosis, and/or priming of eosinophils (IL-5) and mast cells (IL-9); and IL-17 plays an important role in neutrophil homeostasis, recruitment, and activation.

These same cytokines alter the function of innate immune APCs (e.g., DCs), regulating the subsequent generation of polarized populations of effector lymphocytes. Polarized effector lymphocytes also regulate DC function via cell surface ligands that up- or downregulate antigen presentation (e.g., via CD40L and CTLA-4, respectively).

### Counter-Regulation of Innate Immune Responses

As might be expected, the adaptive immune system plays a role in counter-regulation of innate immune responses. Again this occurs through multiple, often overlapping pathways, including specialized cell types (e.g., Tregs, Th3 cells, regulatory B cells), inhibitory cytokines (e.g., IL-10, TGF-β, IL-4), and inhibitory enzymes (e.g., inducible nitric oxide and IDO – both upregulated by IFN-γ).

## TISSUE-SPECIFIC IMMUNITY: ORGAN-SPECIFIC REGULATION OF IMMUNE RESPONSES

As the increasingly convoluted descriptions of immune activation and regulation found above should make clear, distinctions between innate and adaptive immunity are, in many ways, artificial. The immune system acts in a coordinated fashion to respond to insults. Reductive distinctions between various facets of the immune system have theoretical and experimental utility – but, in the end, such facets can only be understood in the wider context of the immune system as a whole.

Another critical, contextual piece to be kept in mind is that immune responses occur in specific tissues. The biologically appropriate class of immune response varies not only by type of insult, but also by tissue. For example, the lung represents a very large (approximately a tennis court in surface area), very thin (two cells thick) interface with the external environment that is specialized for gas exchange. Constantly exposed to danger in the form of inhaled and aspirated microbes and microbial products, the normal function of the lung can rapidly be compromised by the destructive power of inflammatory responses. It is therefore not surprising that the lung, normally kept sterile, is equipped with a prodigious array of innate immune mechanisms that allow for constitutive antimicrobial activity in the absence of inflammation; that immune activation, the class of immune activation and the amplitude of immune activation is tightly controlled in the lung; and that the lung is marked by particularly strong immune counter-regulation. On the other hand, the gut mucosa is bathed by commensal flora in numbers that exceed the total number of cells in the host organism. Failing to respond to commensal flora, responding appropriately to pathogenic microbes, and doing the latter without functional derangement of gut function are obvious, important problems for the gut mucosal immune system. Again, gut immune responses (often marked by IL-10 and TGF-β expression and secretory IgA production) are tightly controlled in terms of activation, class, and amplitude. Similar biological considerations come to the fore in sites such as the eye, brain, testis, and placenta, where functional and structural constraints dictate the biological utility of specific classes of immune response. Matzinger has pointed out that immunologists have traditionally talked about such sites as being "immunologically privileged," because the sorts of immune responses that immunologists were used to measuring were hard to generate in these sites. But, in fact, this represents not a failure of immune activation, but immune class deviation; these sites mount quite effective, biologically appropriate immune responses.

Our knowledge of how specific organs and tissues "predispose" to certain classes of immune response remains somewhat spotty. In part, this is likely a function of tissue-specific parenchymal recruitment and "education" of DC subtypes (master regulators of immune class specification), tissue macrophage populations (long known to have dramatic, organ-specific functional heterogeneity), and innate lymphocyte populations (including NKT cells in the liver, innate B cells in the peritoneum, and γ/δ T cells in the gut). In turn, organ-specific innate immune cell repertoires (including, quite importantly, parenchymal cells such as airway epithelia) are likely to regulate the induction

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(and resolution) of organ- (and class-) specific immune responses, both innate and adaptive.

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